Coming out from the shadows: facultative slave-making ants reveal their chemical identity during colony development

Behavioral Ecology and Sociobiology

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Contents

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1 Introduction

This document presents graphical results generated as output from analytic workflows written in *R* language and available in the public GitHub repository - https://github.com/TomVuod/sangchem/tree/supp.

2 Species discrimination and prediction

To classify the CHC samples collected from mixed colonies, we need a measure which positions a sample between the *F. sanguinea* and *F. fusca* CHC profiles. Therefore, using `mixOmics` R package [Le Cao et al., 2025], we performed a discriminant analysis training the model on the chemical data collected from the filed colonies in the earlier study [Włodarczyk and Szczepaniak, 2017]. The model was subsequently used to predict identity of new samples.

2.1 Species markers

To figure out the importance of each input variable in the prediction of the species identity made by discriminant model, we calculated the vector of weights using the following formula [Tenenhaus, 1998, Rohart et al. [2017]].

$$\mathbf{W}(\mathbf{P}^\top\mathbf{W})^{-1}\mathbf{c}$$

$$\mathbf{P} = \mathbf{X}\mathbf{V}$$

$$\mathbf{c} = \mathbf{V}^\top\mathbf{y},$$

where
$\mathbf{W}$ is the matrix of loading vectors with the number of rows corresponding to the number of input variables and the number of columns corresponding to the number of the latent components
$\mathbf{V}$ is the matrix of the coordinates of each observation on latent components
$\mathbf{X}$ is the input data matrix $\mathbf{y}$ is the vector of zero-centered one-hot encoded class labels of the samples for a selected class (here, *F. sanguinea* sample)

To project the weights assigned to principal components onto the original variables one has to reverse data transformation. Technically, this is achieved by multiplication by pseudoinverse of the rotation matrix produced by PCA. We use pseudoinverse since the rotation matrix has been reduced by removing principal components with low variance.

$$\mathbf{R}^+\mathbf{W}(\mathbf{P}^\top\mathbf{W})^{-1}\mathbf{c},$$

where $\mathbf{R}^+$ denotes the pseudoinverse of rotation matrix from PCA.

Table S1 : List of the peaks with the species identity score indicating their importance as a marker of *F. fusca* or *F. sanguinea* samples.

Peak ID	Compound	Marker score	fusca marker	sanguinea marker
15	4-MeC24	-0.0255545	TRUE	FALSE
60	13,17-diMeC29	-0.0253947	TRUE	FALSE
10	3-MeC23	-0.0199077	TRUE	FALSE
6	11-Me-C23 + 9-Me-C23	-0.0187827	TRUE	FALSE
32	14-; 10-MeC26 + 3,7,11-triMeC25	-0.0181734	TRUE	FALSE
12	3,13-; 3,11-; 3,9-; 3,7-diMeC23	-0.0176211	TRUE	FALSE
14	6-MeC24	-0.0174156	TRUE	FALSE
7	7-Me-C23	-0.0171928	TRUE	FALSE
13	12-; 11-; 10-; 9-; 8-MeC24	-0.0170398	TRUE	FALSE
69	C31	-0.0169988	TRUE	FALSE
44	7-MeC27	-0.0168845	TRUE	FALSE
37	7,x-; 5,x-; 10,14-diMeC26	-0.0168292	TRUE	FALSE
31	3,13-diMeC25	-0.0164987	TRUE	FALSE
19	4,12-diMeC24	-0.0160872	TRUE	FALSE
11	C24	-0.0151770	TRUE	FALSE
22	13-; 11-; 9-MeC25	-0.0150263	FALSE	FALSE
49	7,11-diMeC27	-0.0145556	FALSE	FALSE
5	C23	-0.0145410	FALSE	FALSE
33	6-MeC26	-0.0145371	FALSE	FALSE
27	7,11-diMeC25 + 3-MeC25	-0.0141446	FALSE	FALSE
55	C29ene	-0.0120979	FALSE	FALSE
40	4,12-diMeC26	-0.0117432	FALSE	FALSE

Table S1 : List of the peaks with the species identity score indicating their importance as a marker of *F. fusca* or *F. sanguinea* samples. *(continued)*

Peak ID	Compound	Marker score	fusca marker	sanguinea marker
21	4,12,16-triMeC24	-0.0115601	FALSE	FALSE
65	3,7-; 3,9-; 3,11-diMeC29	-0.0109379	FALSE	FALSE
3	C23-ene	-0.0105481	FALSE	FALSE
8	5-Me C23	-0.0099939	FALSE	FALSE
57	4,8,12-triMeC28	-0.0097465	FALSE	FALSE
42	4,8,14-MeC26	-0.0092601	FALSE	FALSE
26	11,15-; 9,13-diMeC25	-0.0074051	FALSE	FALSE
63	C30ene	-0.0073431	FALSE	FALSE
34	5-MeC26	-0.0072263	FALSE	FALSE
53	14-; 12-; 10-MeC28	-0.0059975	FALSE	FALSE
16	10,14-diMeC24	-0.0049985	FALSE	FALSE
56	C29	-0.0026256	FALSE	FALSE
24	7-MeC25	-0.0024319	FALSE	FALSE
58	15-; 13-; 11-; 9-; 7-MeC29	-0.0017646	FALSE	FALSE
23	9-MeC25	-0.0006146	FALSE	FALSE
20	C25	-0.0002985	FALSE	FALSE
77	C33ene	0.0000217	FALSE	FALSE
43	13-MeC27	0.0000912	FALSE	FALSE
51	C28 + 3,15-diMeC27	0.0002801	FALSE	FALSE
47	9,17-; 9,15-diMeC27	0.0027638	FALSE	FALSE
59	11,17-diMeC29	0.0032268	FALSE	FALSE
66	14-; 13-; 12-; 11-MeC30	0.0037694	FALSE	FALSE
48	9,13-diMeC27	0.0051902	FALSE	FALSE
28	5,13-diMeC25	0.0054108	FALSE	FALSE
9	9,13-diMeC23	0.0059293	FALSE	FALSE
4	C23-ene	0.0060814	FALSE	FALSE
80	11,21-; 11,19-diMeC33	0.0091259	FALSE	FALSE
36	10,16-; 10,14-diMeC26	0.0094887	FALSE	FALSE
52	3,9-; 3,7-diMeC27	0.0095619	FALSE	FALSE
54	8,12,16-triMeC28	0.0097784	FALSE	FALSE
18	C25-ene	0.0098478	FALSE	FALSE
72	13,19-;13,17-diMeC31	0.0098522	FALSE	FALSE
30	C26	0.0103193	FALSE	FALSE
62	5,9-; 5,11-; 5,13-diMeC29	0.0103978	FALSE	FALSE
75	x-Me_C32	0.0106378	FALSE	TRUE
79	x-Me_C32	0.0131912	FALSE	TRUE
46	11,15-diMeC27	0.0143588	FALSE	TRUE
74	(di)MeC31, mix	0.0145589	FALSE	TRUE
70	15-; 13-; 11-; 9-MeC31	0.0150606	FALSE	TRUE
45	5-MeC27	0.0154463	FALSE	TRUE
35	4-MeC26	0.0156798	FALSE	TRUE
76	13,17-DiMe_C32 + x,y-DiMe_C32	0.0157185	FALSE	TRUE
17	C25-ene	0.0177937	FALSE	TRUE
71	11,19-; 11,17-; 11,15-diMeC31	0.0180838	FALSE	TRUE
41	C27	0.0196690	FALSE	TRUE
38	C27-ene + 6,12-diMeC26	0.0222253	FALSE	TRUE

Table S1 : List of the peaks with the species identity score indicating their importance as a marker of *F. fusca* or *F. sanguinea* samples. *(continued)*

Peak ID	Compound	Marker score	fusca marker	sanguinea marker
67	(di)MeC30 (mix)	0.0296362	FALSE	TRUE
25	5-MeC25	0.0309633	FALSE	TRUE
50	C28ene	0.0333263	FALSE	TRUE

2.2 Species Identity Score

Ants in the mixed colonies exchanged their CHC to produce a hybrid chemical identity. We developed a method to assign each sample a value measuring similarity to one or the other species. It is based on a discriminant model that predicts the class of new samples using `predict` method from `mixOmics` R package. The input data are processed in the same way as the training set, i.e. central-log transformation is followed by projection into principal component space. In the latter step, the same rotation matrix and scaling vector is applied as those used for the training set are applied. This is achieved by using `predict` method on the object of `prcomp` class.

2.2.1 Predicted species of different categories of ants as a function of the proportion of *F. sanguinea* ants in a colony

The Species Identity Score was computed for *F. sanguinea* and *F. fusca* samples from mixed colonies and used as a response variable in linear mixed models. As a fixed effect we used the proportion of *F. sanguinea* workers among all workers in a colony. Colony ID and sampling occasion were incorporated as random effects.

In the following model specifications (1|random_factor) denotes random intercept (separate for each level of the random factor) and ((1|random_factor_1:random_factor_2)) denotes random intercepts generated from the combinations of two factors. β_0 and ϵ denote intercept and error term, respectively. Included are summary reports, results of the Shapiro-Wilk test of nornalizty of conditional residuals, as well as diagnostic plots and tests generated with the use of `DHARMa` R package[Hartig, 2024]. The response variable often needed to be transformed to meet model assumptions. Random terms with no variance have been dropped.

2.2.1.1 Mature *F. sanguinea*

$$\sqrt{\text{Species\_Identity\_Score+1}} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

where Species_Identity_Score denotes the Species Identity Score of the mature *F. sanguinea* workers.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(predicted_species + 1)) ~ sang_prop + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 22.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.39156 -0.43663  0.06261  0.57841  1.64975
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
```

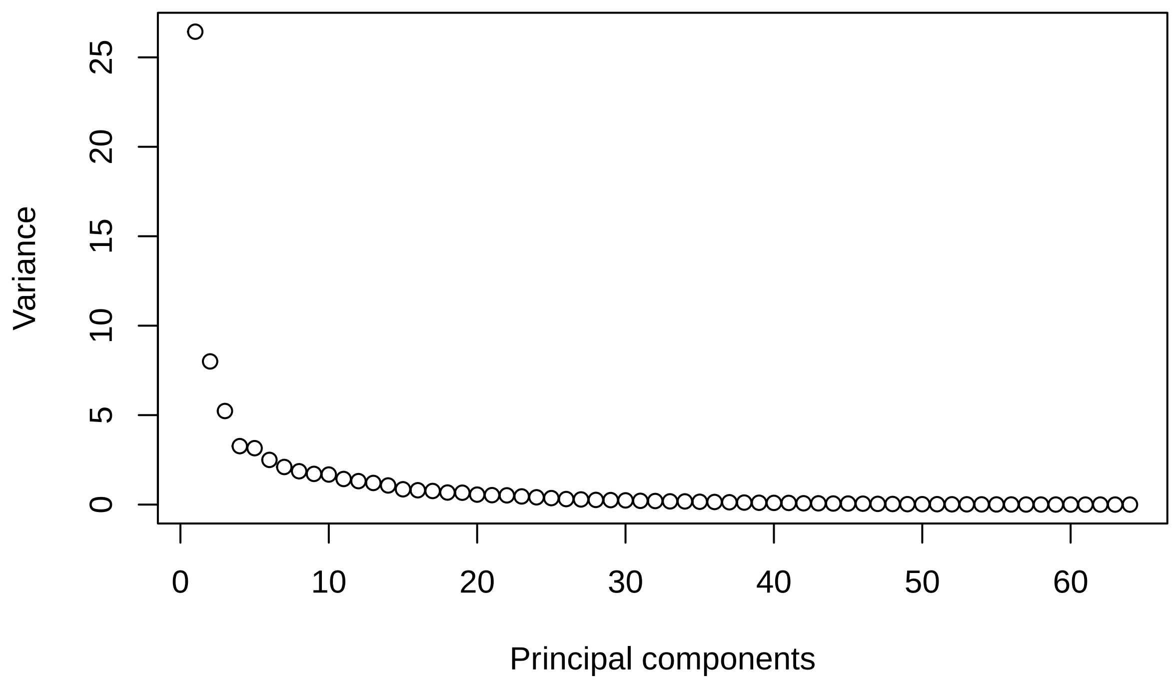


Figure S1 : Variance of each principal compnent. Data was fed into the discriminat model to find features separating both species.

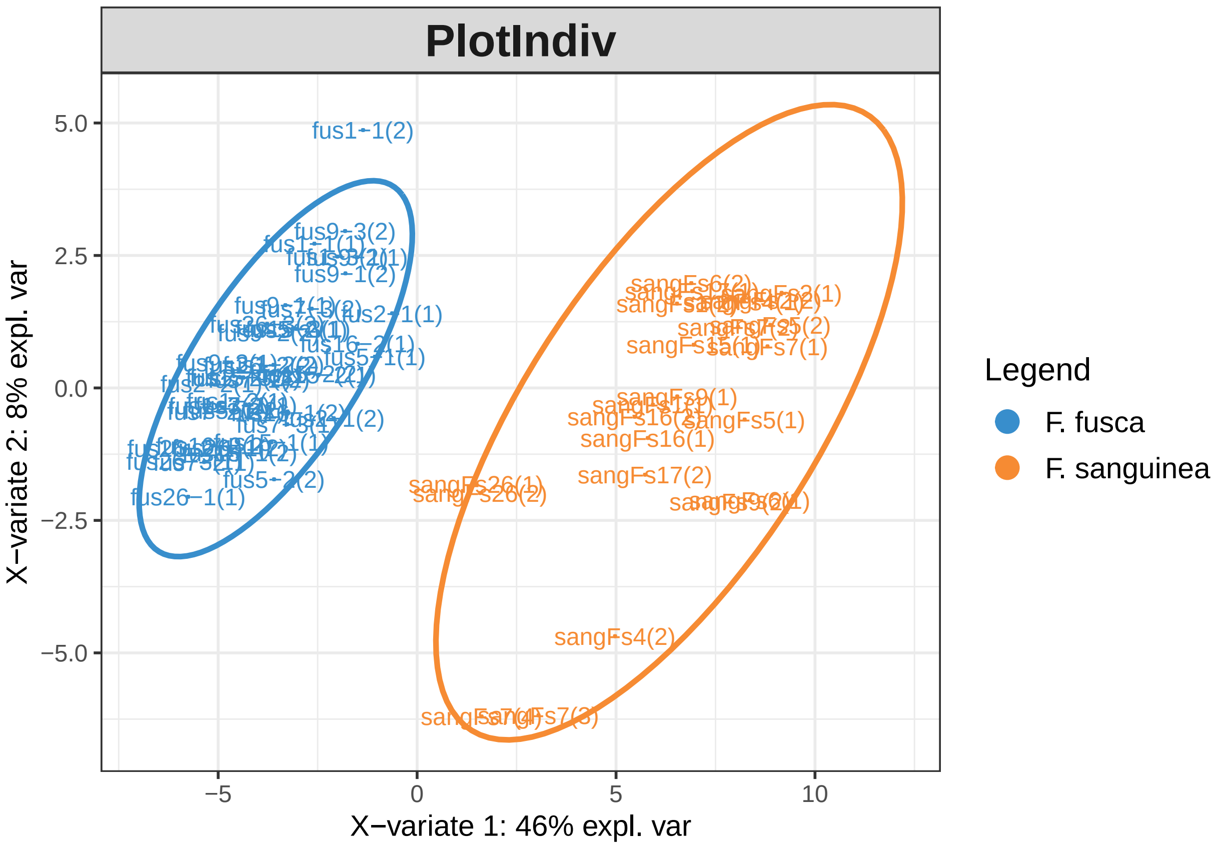


Figure S2 : Distribution of the samples projected onto two first latent components before optimizing the number of model paramaters.

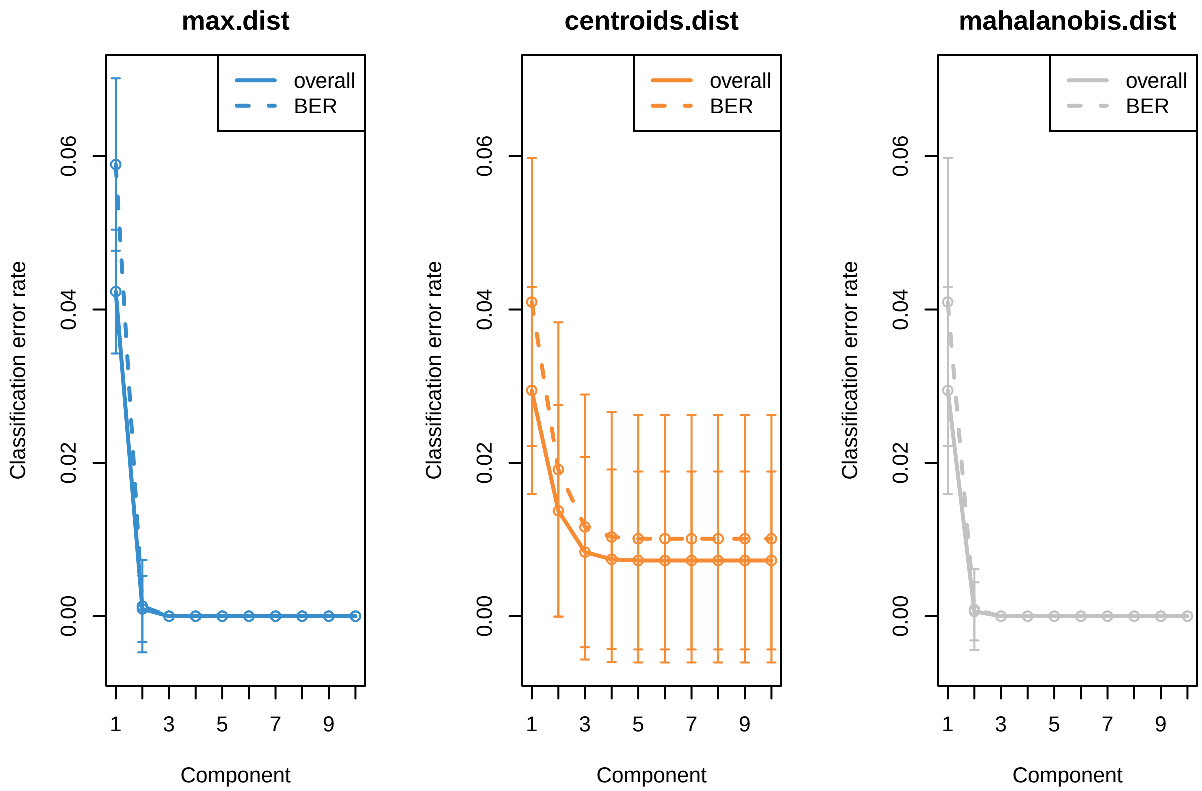


Figure S3 : Results of the performance test of the discriminant model with different number of the latent components. For details see the documunetation of R mixOmics package.

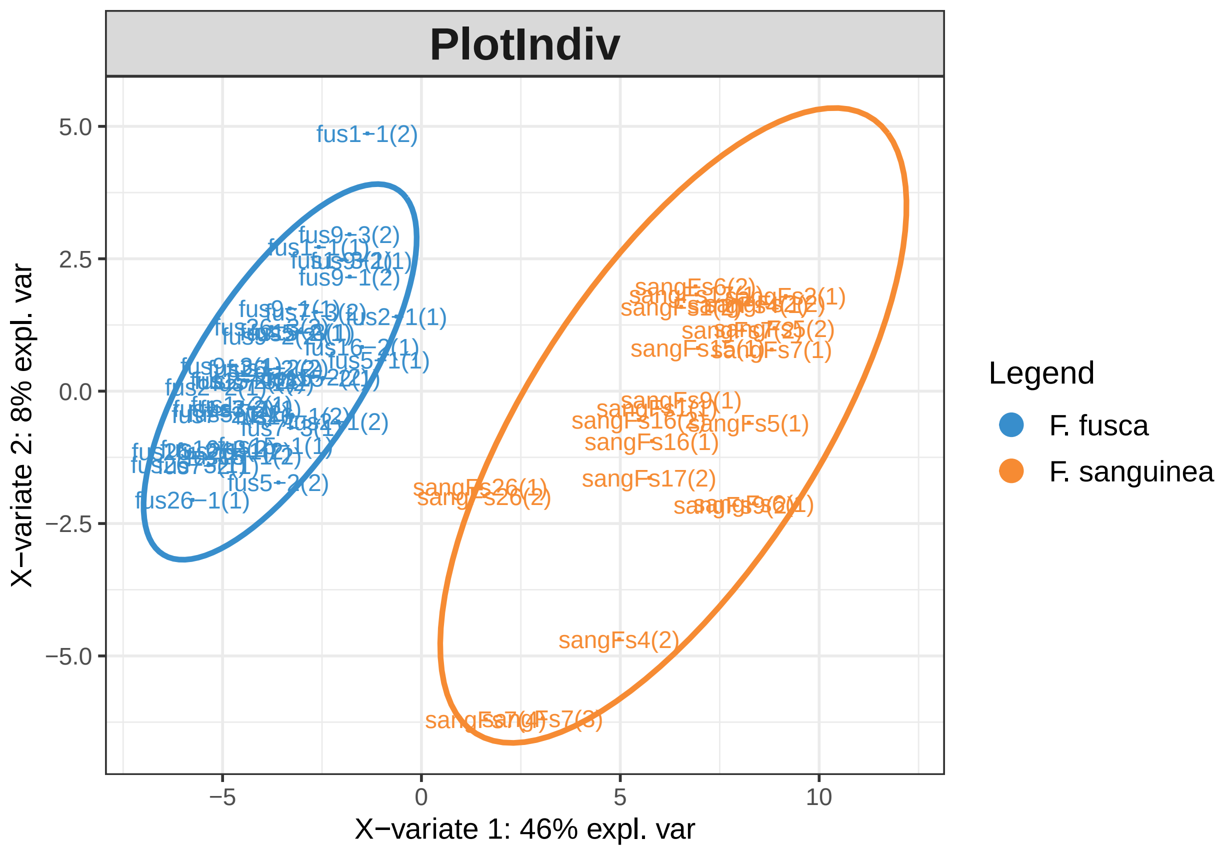
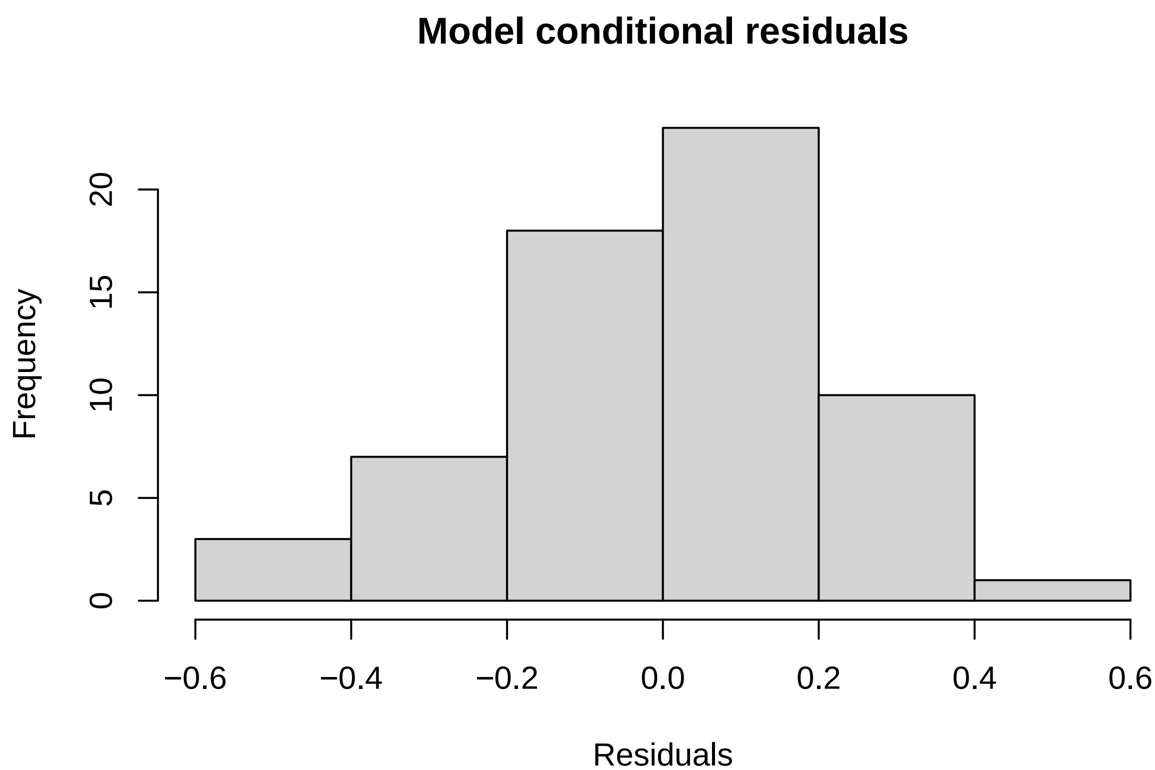
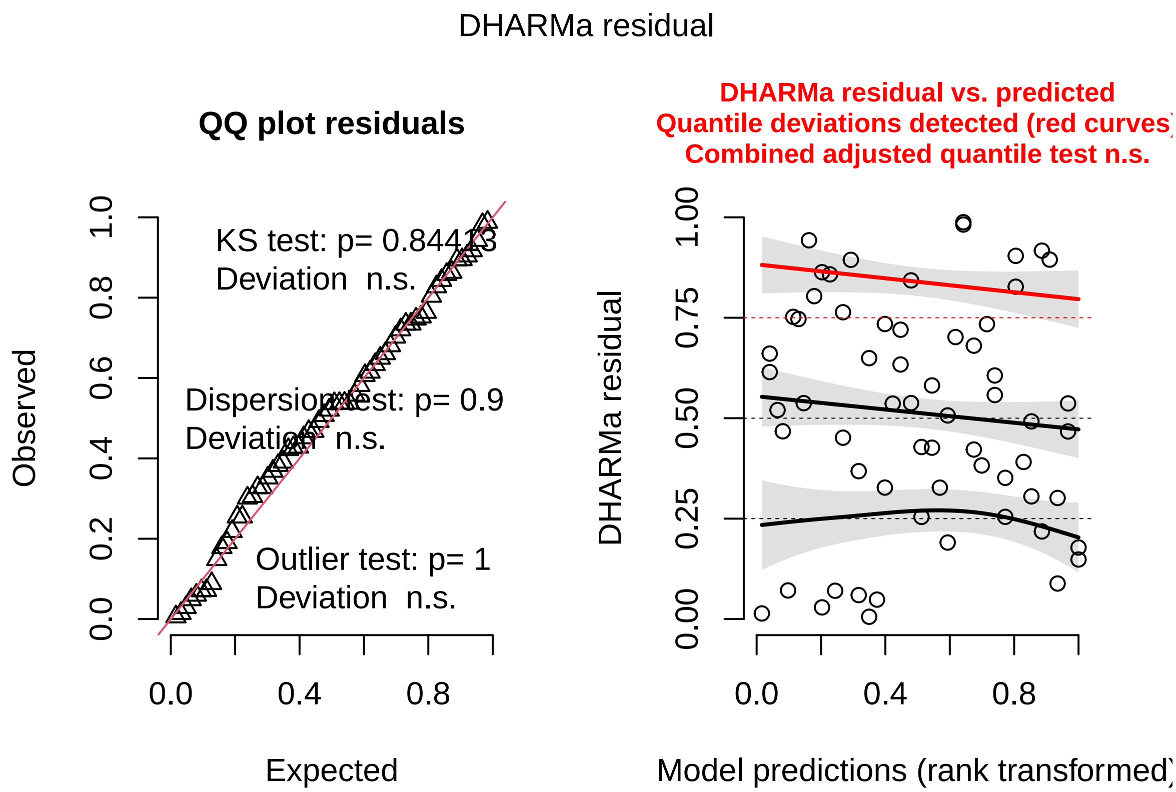


Figure S4 : Projection of the samples in onto two first discriminant analysis components after model tuning.

```
##  colony:census_date (Intercept) 0.01917  0.1385
##  Residual                        0.06029  0.2455
## Number of obs: 62, groups:  colony:census_date, 42
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  0.96117    0.07083 42.22312  13.570  < 2e-16 ***
## sang_prop    1.00289    0.13239 39.12506   7.575 3.46e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.841
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.9752, p-value = 0.2413
```





2.2.1.2 *F. fusca*

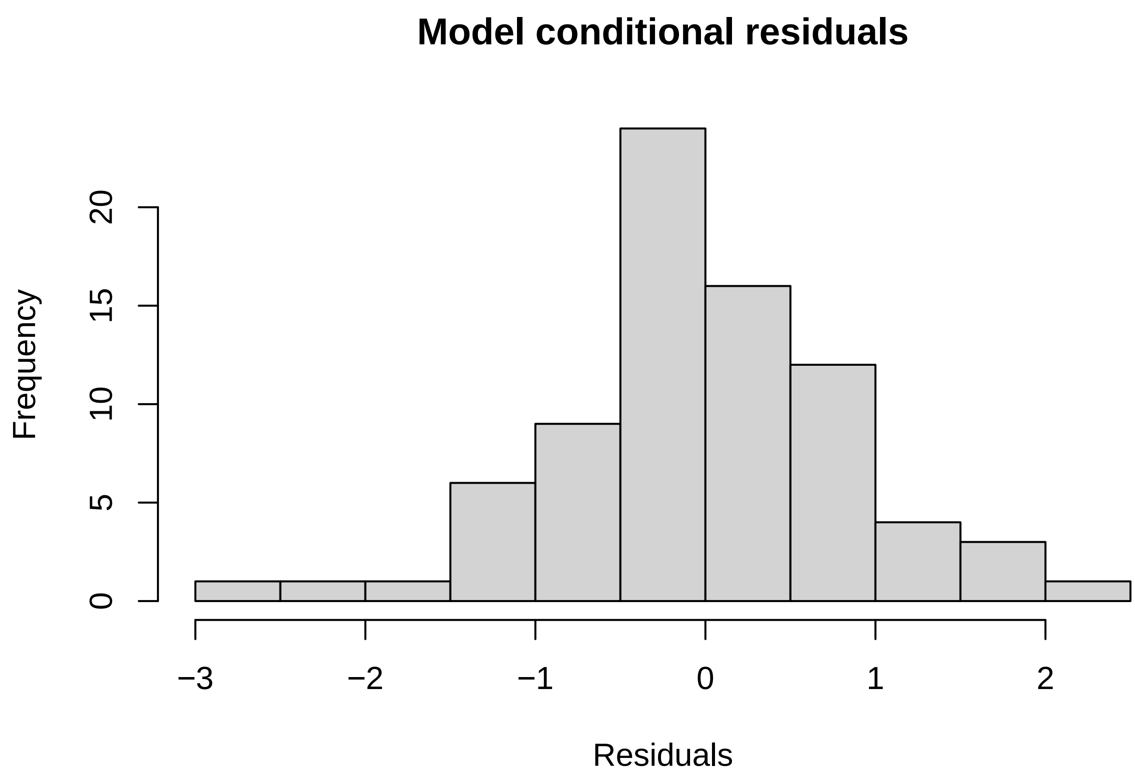
$$\text{Species\_Identity\_Score} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

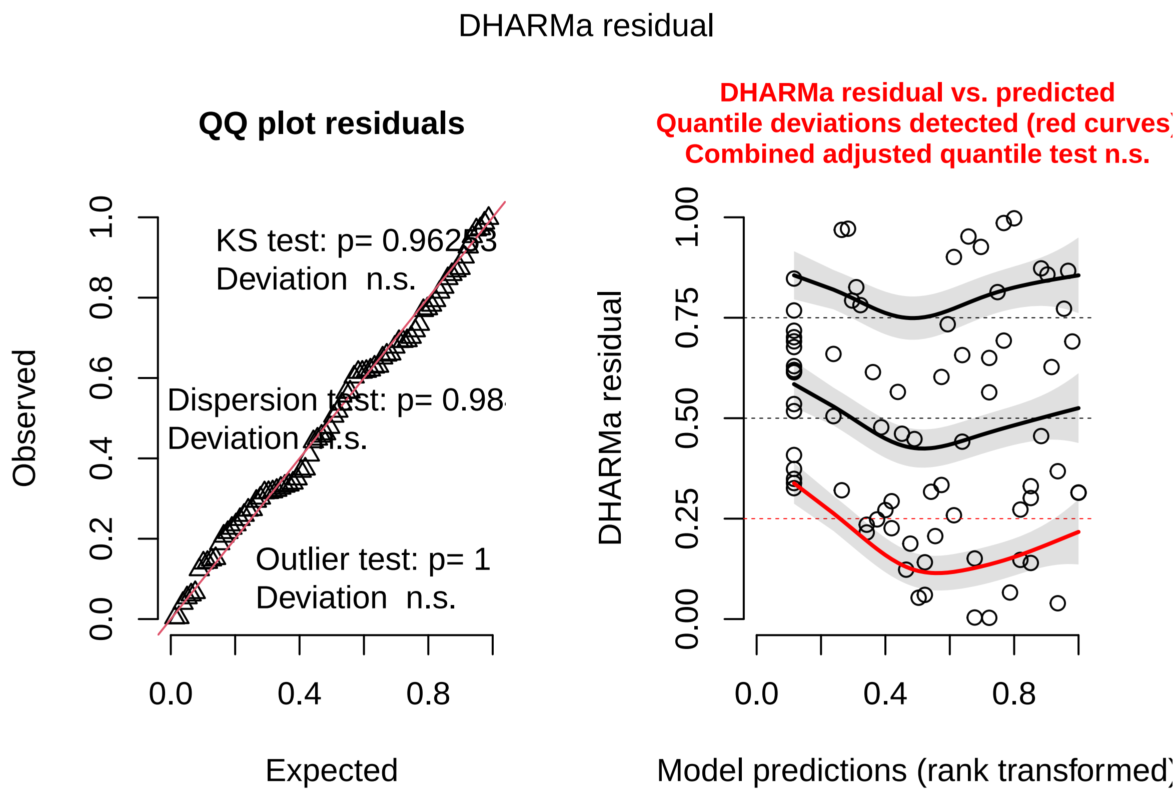
where Species_Identity_Score denotes the Species Identity Score of the mature *F. fusca* workers.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: predicted_species ~ sang_prop + (1 | colony) + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 240.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.73942 -0.41900 -0.07327  0.52549  2.41878
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.2073   0.4553
##  colony             (Intercept) 0.1479   0.3846
##  Residual                       0.9773   0.9886
## Number of obs: 78, groups:  colony:census_date, 55; colony, 20
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)  -1.5743     0.2110 42.6506   -7.46 2.93e-09 ***
## sang_prop     3.3945     0.4606 36.1850    7.37 1.05e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.661
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.9847, p-value = 0.4753
```





2.2.1.3 Callow *F. sanguinea*

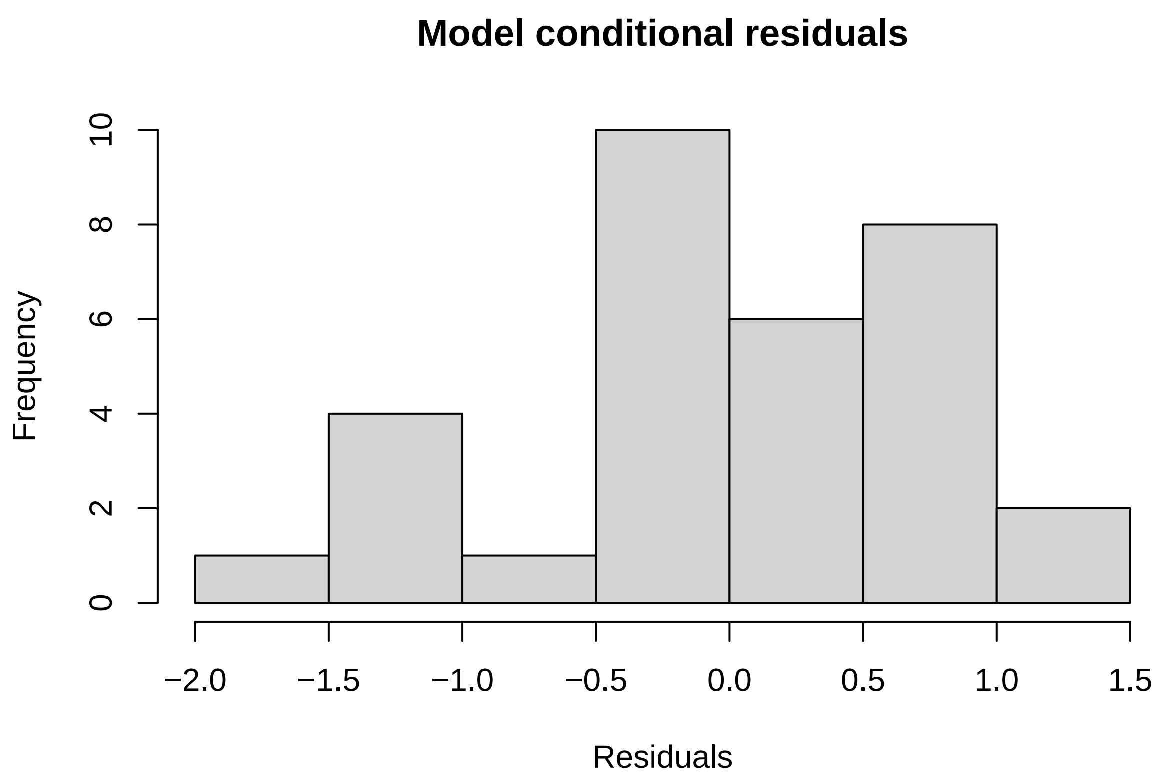
$$\text{Species\_Identity\_Score} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony}) + \epsilon,$$

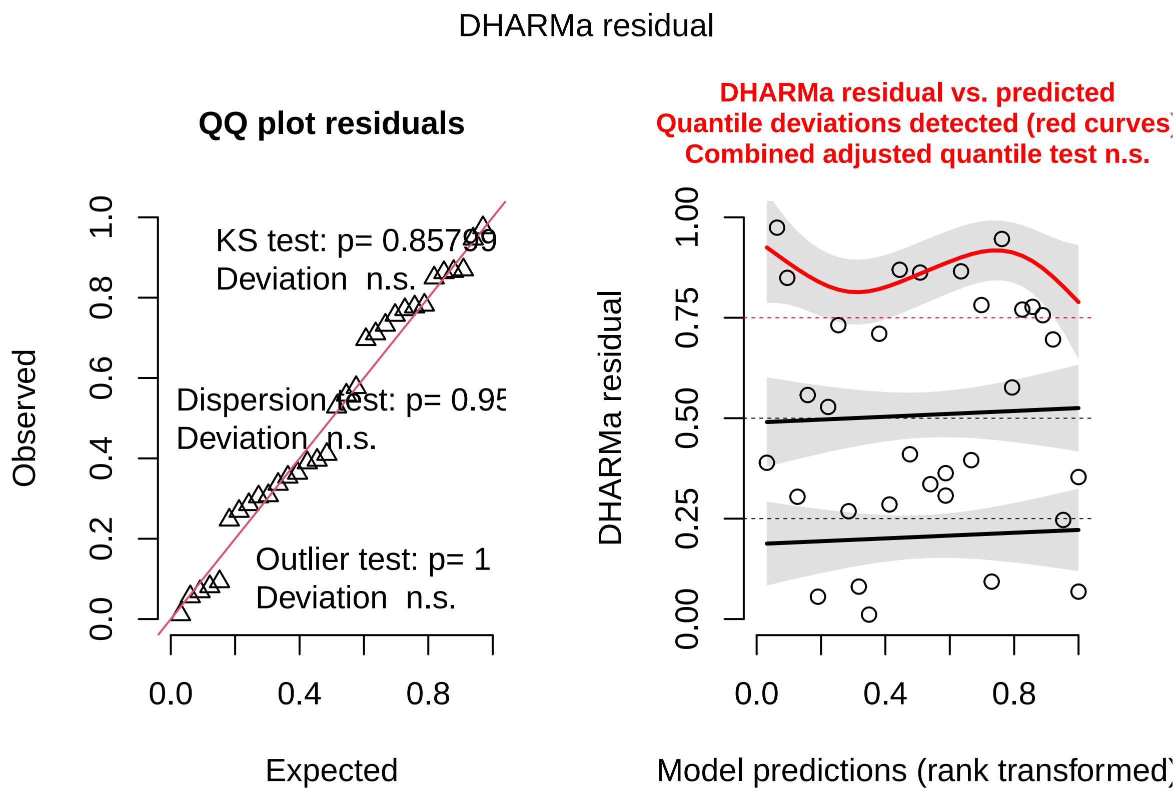
where Species_Identity_Score denotes the Species Identity Score of the callow *F. sanguinea* workers.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: predicted_species ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 76.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.25648 -0.51742 -0.08274  0.69855  1.88538
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.01975  0.1406
##  Residual             0.62134  0.7883
## Number of obs: 32, groups:  colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.9408     0.2530 28.3318  -3.719 0.000877 ***
```

```
## sang_prop     2.5681     0.4727 29.6945   5.433 7.07e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.821
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.97706, p-value = 0.7106
```

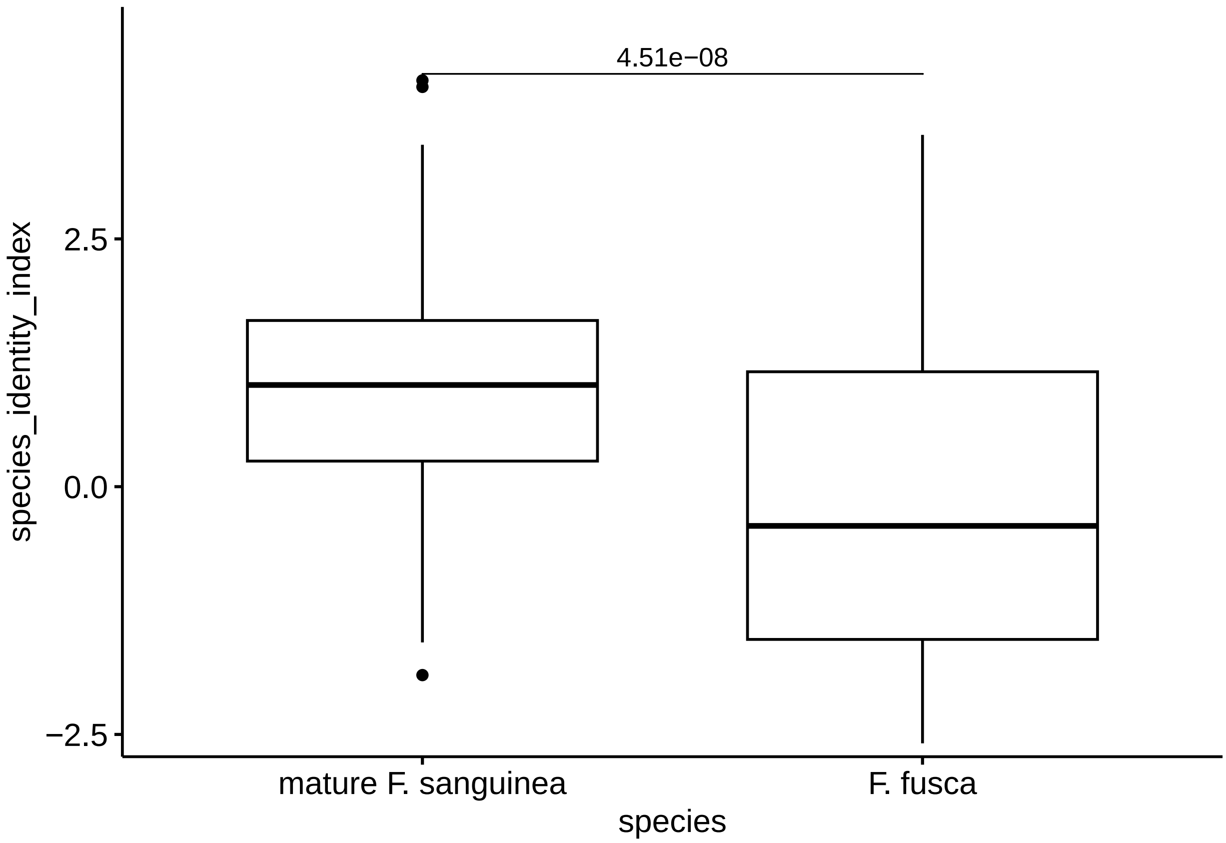




2.2.2 Difference in Species Identity Score between mature *F. sanguinea* ants and *F. fusca* slaves

Since the linear model does not meet one of the diagnostics criteria, Wilcoxon paired test was applied.

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  model_input$predicted_species and model_input$slave_species_index
## V = 1910, p-value = 4.507e-08
## alternative hypothesis: true location shift is not equal to 0
```



2.2.3 Difference in Species Identity Score between callow *F. sanguinea* ants and *F. fusca* slaves

$$\text{SIS\_diff} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony}) + \epsilon,$$

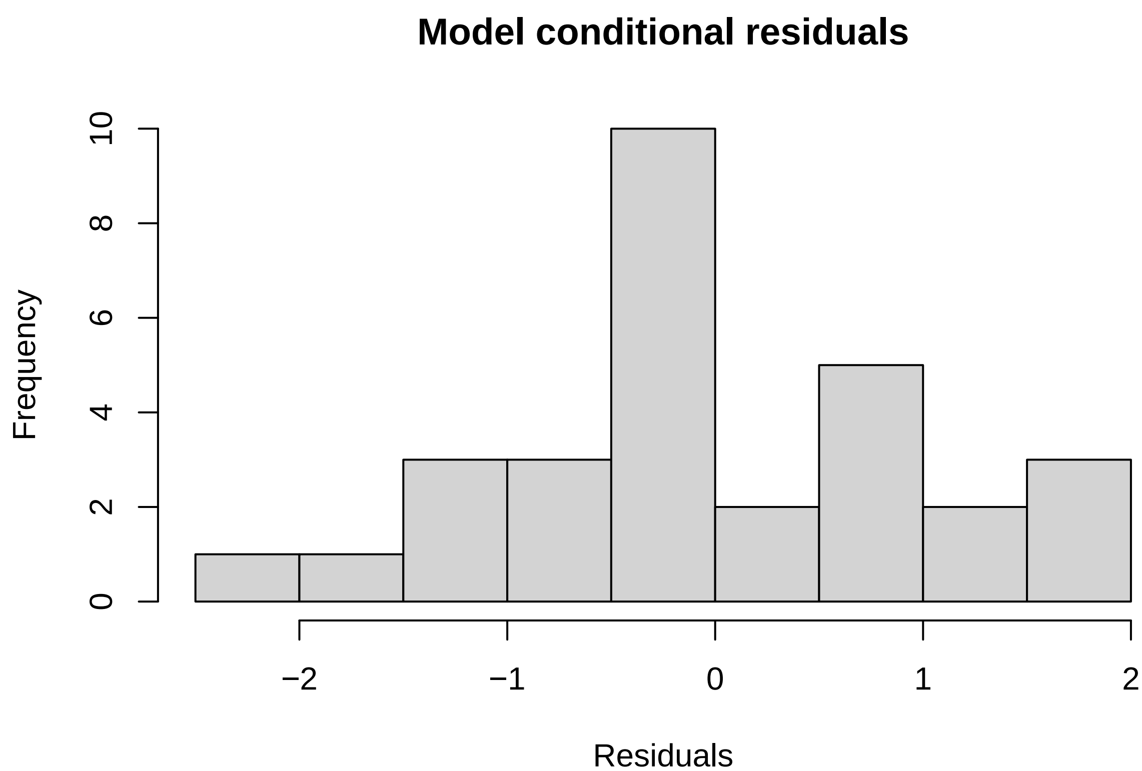
where SIS_diff denotes the difference in Species Identity Score between the mature *F. sanguinea* and *F. fusca* sampled at the same time and from the same colony. Replicated samples were averaged before the calculations.

Since the linear model does not meet one of diagnostics criterion we will apply Wilcoxon paired test.

```
##
## Call:
## lm(formula = index_diff ~ sang_prop, data = model_input)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1745 -0.6534 -0.1962  0.7732  1.9023
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.3914     0.3492   1.121    0.272
## sang_prop    -0.4235     0.7201  -0.588    0.561
##
## Residual standard error: 1.065 on 28 degrees of freedom
##   (2 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.0122, Adjusted R-squared:  -0.02308
## F-statistic: 0.3458 on 1 and 28 DF,  p-value: 0.5612
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.97627, p-value = 0.7202
```

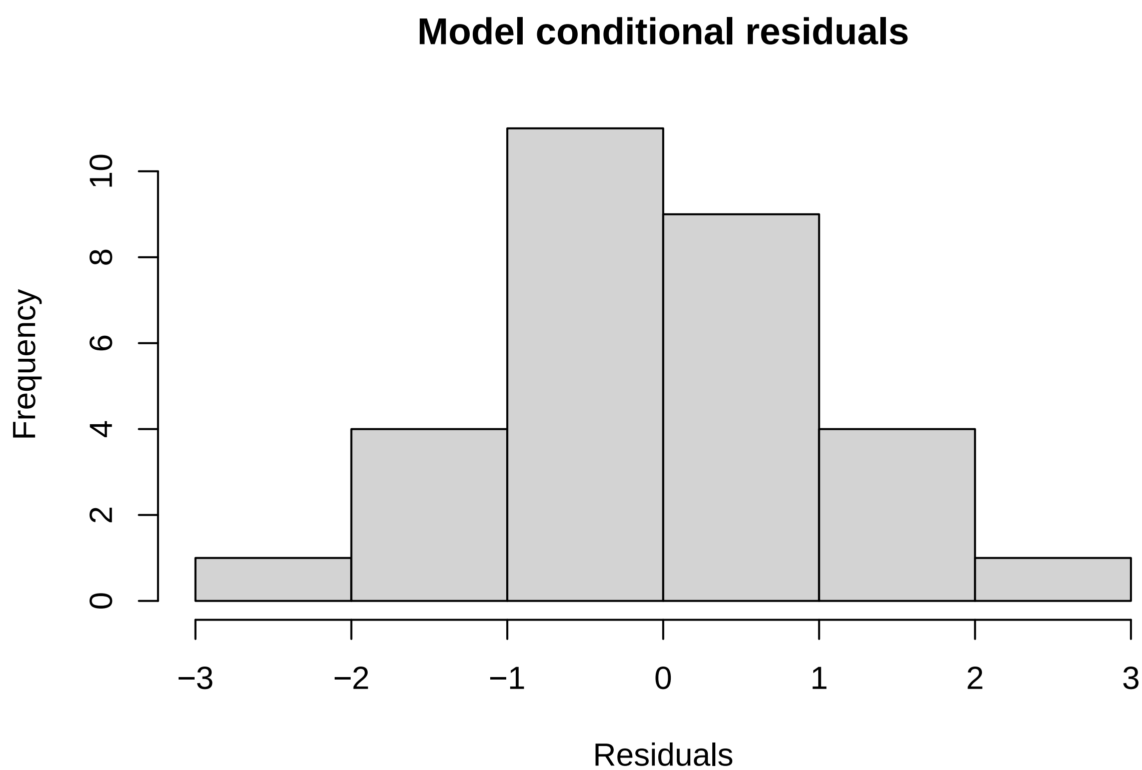


2.2.4 Difference in Species Identity Score between callow and mature *F. sanguinea* ants.

```
##
## Call:
## lm(formula = index_diff ~ sang_prop, data = model_input)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.3512 -0.4805 -0.1439  0.6242  2.1423
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.92662    0.33558  -2.761     0.01 *
## sang_prop    0.07754    0.61392   0.126     0.90
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.002 on 28 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.0005694,  Adjusted R-squared:  -0.03512
## F-statistic: 0.01595 on 1 and 28 DF,  p-value: 0.9004

##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99264, p-value = 0.9987
```



3 Finding peaks differentiating callow and mature *F. sanguinea* workers

The procedure identifying species markers was also applied to determine peaks characteristic of *F. sanguinea* callow ants. In discriminant analysis, samples were classified into one of the three groups: callow *F. sanguinea*, adult *F. sanguinea*, or adult *F. fusca*.

3.1 Marker indentification

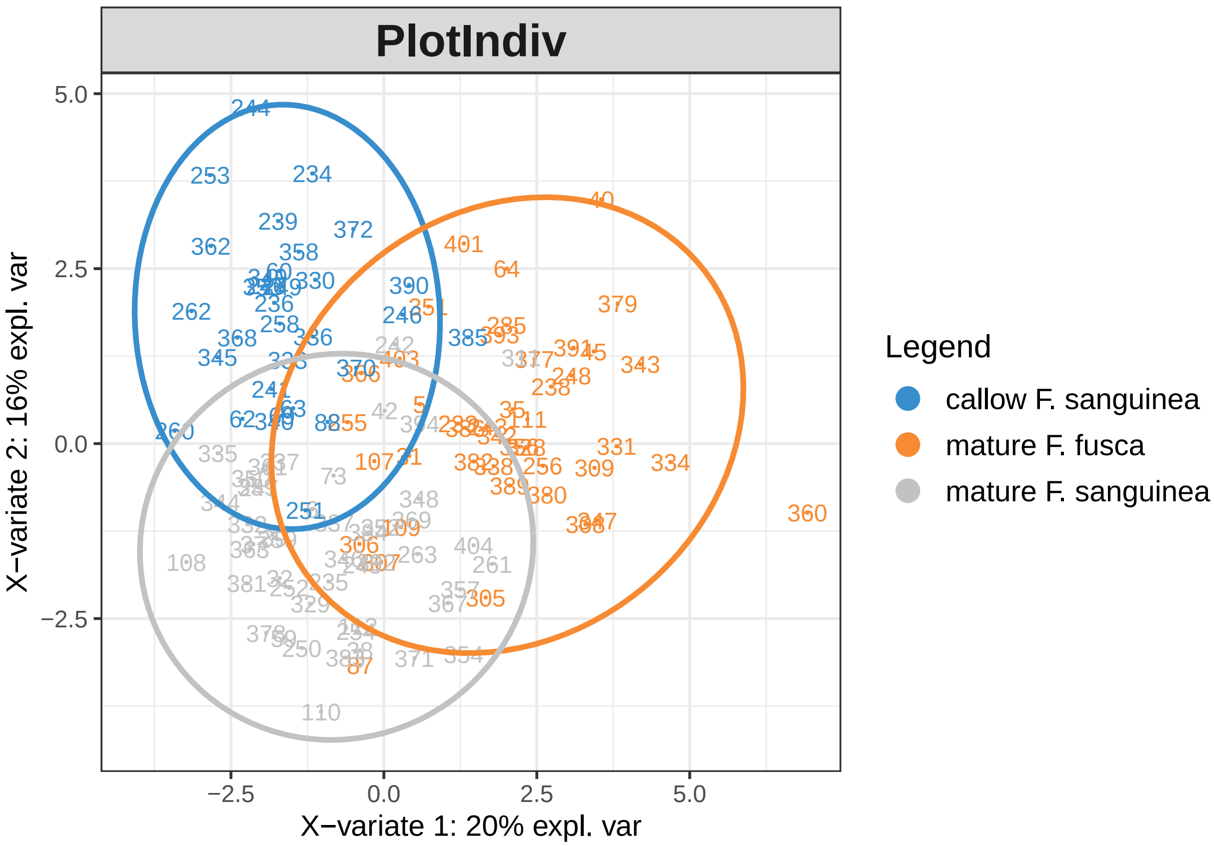


Figure S5 : Distribution of the samples projected onto two first latent components before optimizing the number of model paramaters.

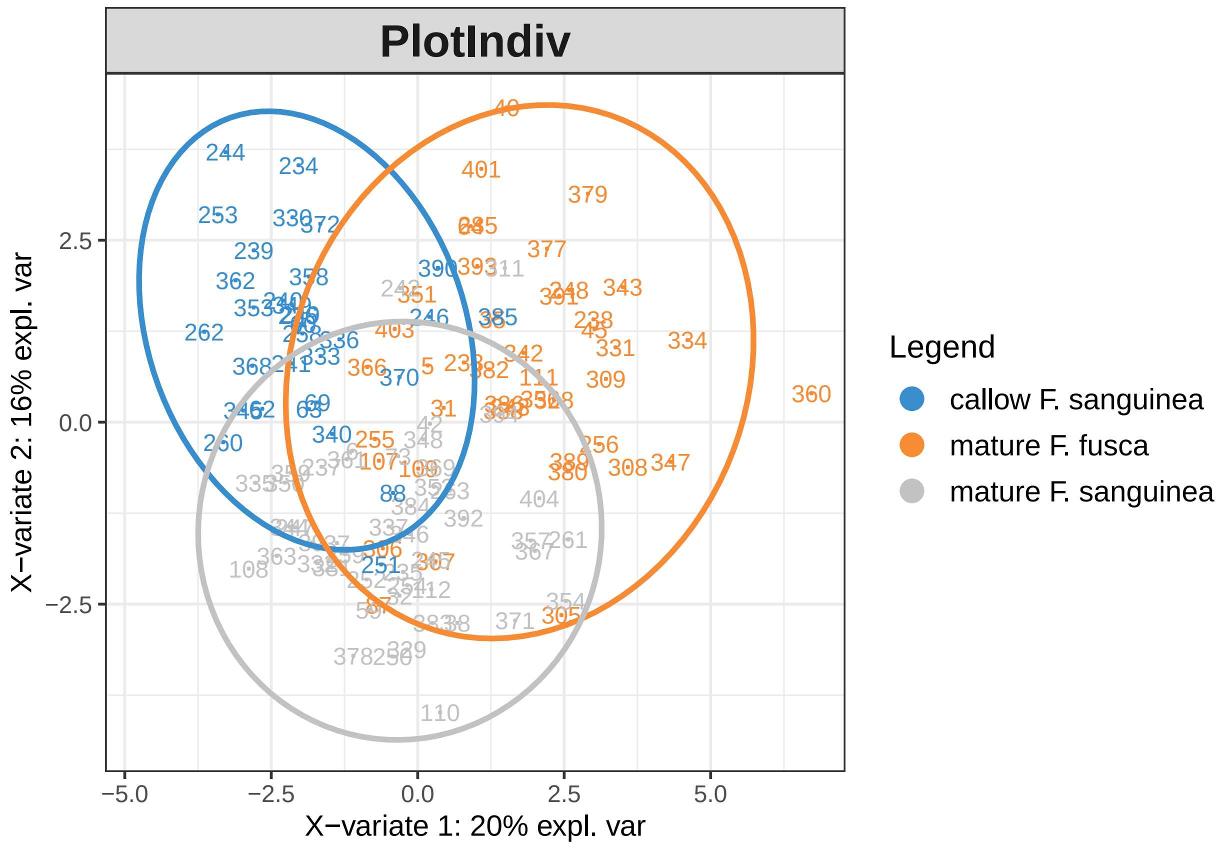


Figure S6 : Projection of the samples in onto two first discriminant analysis components after model tuning.

Table S2 : List of the peaks with the score indicating their importance as a marker of mature *F. fusca*, mature *F. sanguinea*, or callow *F. sanguinea* samples.

Peak ID	Compound	Callow score	Callow marker	Mature score	Mature marker
31	3,13-diMeC25	0.0416903	TRUE	-0.0178266	FALSE
24	7-MeC25	0.0386839	TRUE	-0.0327207	FALSE
28	5,13-diMeC25	0.0348565	TRUE	-0.0317241	FALSE
43	13-MeC27	0.0305096	TRUE	-0.0234311	FALSE
32	14-; 10-MeC26 + 3,7,11-triMeC25	0.0296012	TRUE	-0.0373441	FALSE
68	C31ene; 3-MeC30	0.0295308	TRUE	-0.0189449	FALSE
58	15-; 13-; 11-; 9-; 7-MeC29	0.0264321	TRUE	-0.0072612	FALSE
53	14-; 12-; 10-MeC28	0.0244925	TRUE	-0.0172651	FALSE
22	13-; 11-; 9-MeC25	0.0241580	TRUE	-0.0457490	FALSE
13	12-; 11-; 10-; 9-; 8-MeC24	0.0235367	TRUE	-0.0429234	FALSE
10	3-MeC23	0.0233218	TRUE	-0.0332150	FALSE
12	3,13-; 3,11-; 3,9-; 3,7-diMeC23	0.0217166	TRUE	-0.0258548	FALSE
52	3,9-; 3,7-diMeC27	0.0207065	FALSE	-0.0121810	FALSE
27	7,11-diMeC25 + 3-MeC25	0.0199437	FALSE	-0.0195216	FALSE
38	C27-ene + 6,12-diMeC26	0.0184220	FALSE	0.0157725	FALSE
19	4,12-diMeC24	0.0174162	FALSE	-0.0198991	FALSE
49	7,11-diMeC27	0.0161116	FALSE	-0.0162587	FALSE
26	11,15-; 9,13-diMeC25	0.0147629	FALSE	-0.0161852	FALSE
6	11-Me-C23 + 9-Me-C23	0.0145882	FALSE	-0.0286965	FALSE
23	9-MeC25	0.0127650	FALSE	0.0068551	FALSE
7	7-Me-C23	0.0126559	FALSE	-0.0155144	FALSE
47	9,17-; 9,15-diMeC27	0.0123952	FALSE	-0.0148502	FALSE
36	10,16-; 10,14-diMeC26	0.0111900	FALSE	-0.0088143	FALSE
17	C25-ene	0.0111205	FALSE	0.0258040	TRUE
54	8,12,16-triMeC28	0.0073623	FALSE	-0.0072445	FALSE
70	15-; 13-; 11-; 9-MeC31	0.0054970	FALSE	0.0234269	TRUE
3	C23-ene	0.0044117	FALSE	-0.0328012	FALSE
5	C23	0.0040366	FALSE	-0.0384879	FALSE
66	14-; 13-; 12-; 11-MeC30	0.0035717	FALSE	0.0179697	FALSE
45	5-MeC27	0.0006637	FALSE	0.0242265	TRUE
72	13,19-;13,17-diMeC31	0.0002333	FALSE	0.0133734	FALSE
60	13,17-diMeC29	-0.0004252	FALSE	0.0173024	FALSE
61	7,11-; 7,13-; 7,15-diMeC29	-0.0024105	FALSE	0.0053526	FALSE
44	7-MeC27	-0.0043353	FALSE	-0.0217227	FALSE
75	x-Me_C32	-0.0048799	FALSE	0.0234993	TRUE
77	C33ene	-0.0086063	FALSE	0.0217736	FALSE
50	C28ene	-0.0108589	FALSE	0.0394054	TRUE
59	11,17-diMeC29	-0.0115280	FALSE	0.0183653	FALSE
67	(di)MeC30 (mix)	-0.0137091	FALSE	0.0378014	TRUE
76	13,17-DiMe_C32 + x,y-DiMe_C32	-0.0141628	FALSE	0.0414515	TRUE
37	7,x-; 5,x-; 10,14-diMeC26	-0.0143395	FALSE	-0.0207037	FALSE
51	C28 + 3,15-diMeC27	-0.0144112	FALSE	0.0029394	FALSE
79	x-Me_C32	-0.0166740	FALSE	0.0323524	TRUE
48	9,13-diMeC27	-0.0187604	FALSE	0.0029709	FALSE
46	11,15-diMeC27	-0.0188128	FALSE	0.0345456	TRUE

Table S2 : List of the peaks with the score indicating their importance as a marker of mature *F. fusca*, mature *F. sanguinea*, or callow *F. sanguinea* samples. *(continued)*

Peak ID	Compound	Callow score	Callow marker	Mature score	Mature marker
11	C24	-0.0194179	FALSE	-0.0262966	FALSE
78	x-Me-C32	-0.0200349	FALSE	0.0050400	FALSE
56	C29	-0.0272272	FALSE	-0.0143242	FALSE
64	C30	-0.0274913	FALSE	-0.0042224	FALSE
80	11,21-; 11,19-diMeC33	-0.0300790	FALSE	0.0357923	TRUE
71	11,19-; 11,17-; 11,15-diMeC31	-0.0324225	FALSE	0.0406780	TRUE
20	C25	-0.0341250	FALSE	-0.0184365	FALSE
30	C26	-0.0342711	FALSE	0.0024356	FALSE
25	5-MeC25	-0.0360842	FALSE	0.0417524	TRUE
69	C31	-0.0421607	FALSE	0.0033436	FALSE
41	C27	-0.0464208	FALSE	0.0009641	FALSE

3.2 Verifying the performace of the discrimination model

The performance of the discriminant model was evaluated by examining the accuracy of its predictions in a cross-validation test, measured as the area under the receiver operating characteristic curve (AUROC). This procedure was repeated 10^3 times to account for variance resulting from the random splitting of samples into training and validation sets. In a parallel analysis, the model was trained on data with permuted age status of *F. sanguinea* to assess whether the true assignment of samples influenced model performance. Results from both training approaches were paired, and the difference in AUROC was calculated each for pair. The empirical p-value was defined as the proportion of models in which the AUROC after permutation was equal to or greater than the AUROC based on the original data.

The p-value is calculated as proportion of differences with value equal to or less than zero.

All differences are positive, so the p-value is less than 0.001 since the null distribution consists of 1000 values.

4 Changes in CHC amount over time

This section contains a series of linear (mixed) models fitted to examine how the proportion of *F. sanguinea* workers in a colony is associated with the amount of CHC, analyzed as a whole or in subsets. Included are summary reports, results of the Shapiro-Wilk test of normality of conditional residuals, as well as diagnostic plots and tests generated with the use of `DHARMa` R package[Hartig, 2024]. The response variable often needed to be transformed to meet model assumptions. Random terms with no variance have been dropped.

In model specification (1|random_factor) denotes random intercept (separate for each level of the random factor) and ((1|random_factor_1:random_factor_2)) denotes random intercepts generated from the combinations of two factors. β_0 and ϵ denote intercept and error term, respectively.

4.1 Colony growth and body size

n the analyses below, normalized CHC amounts were used as the response variable. Because body shape in both species deviates from an isometric growth pattern (see Section 9), normalizing CHC amounts by the square of head width may introduce bias. To address this, we first examined the relationship between body size (head width) and the proportion of F. sanguinea workers in a colony, and then incorporated the square of head width as an explanatory variable in the linear model.

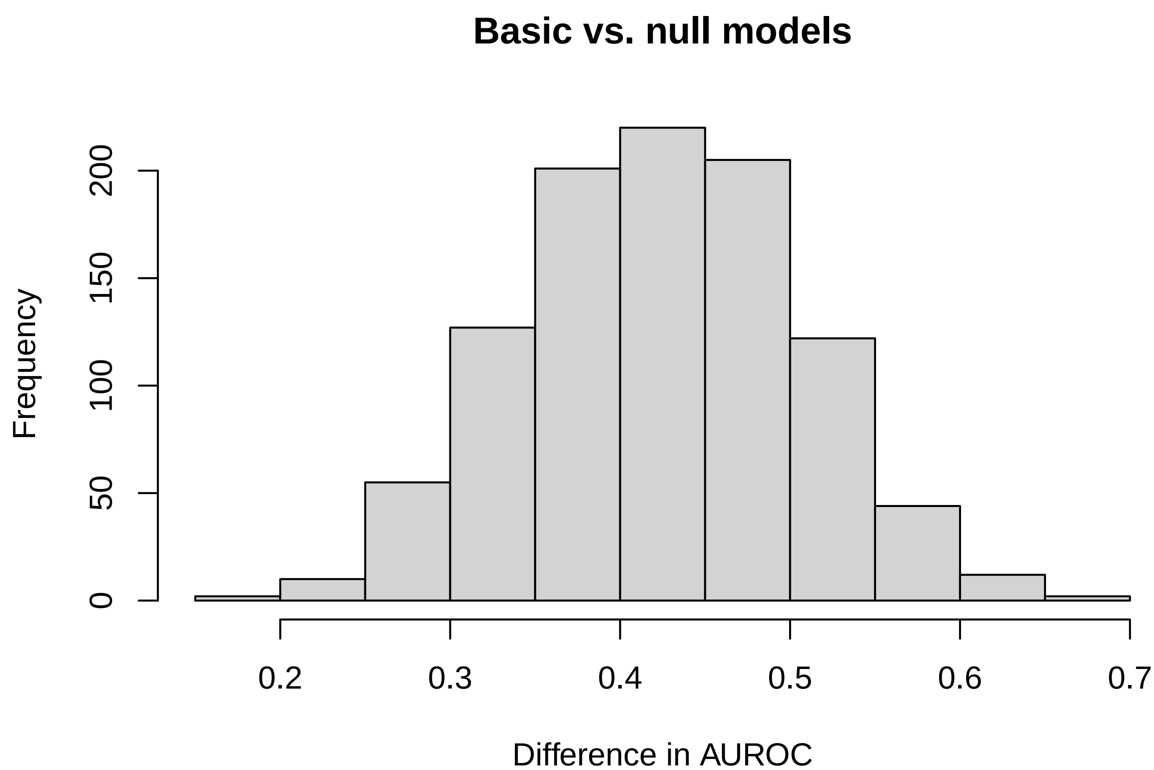
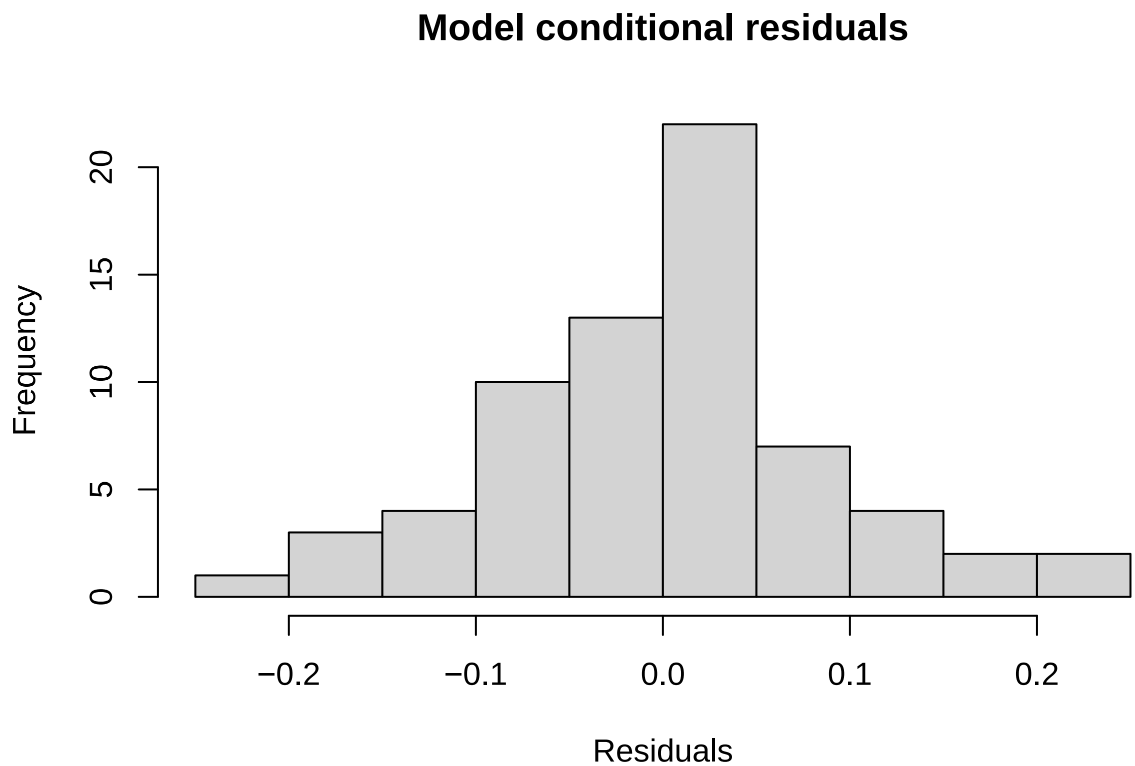


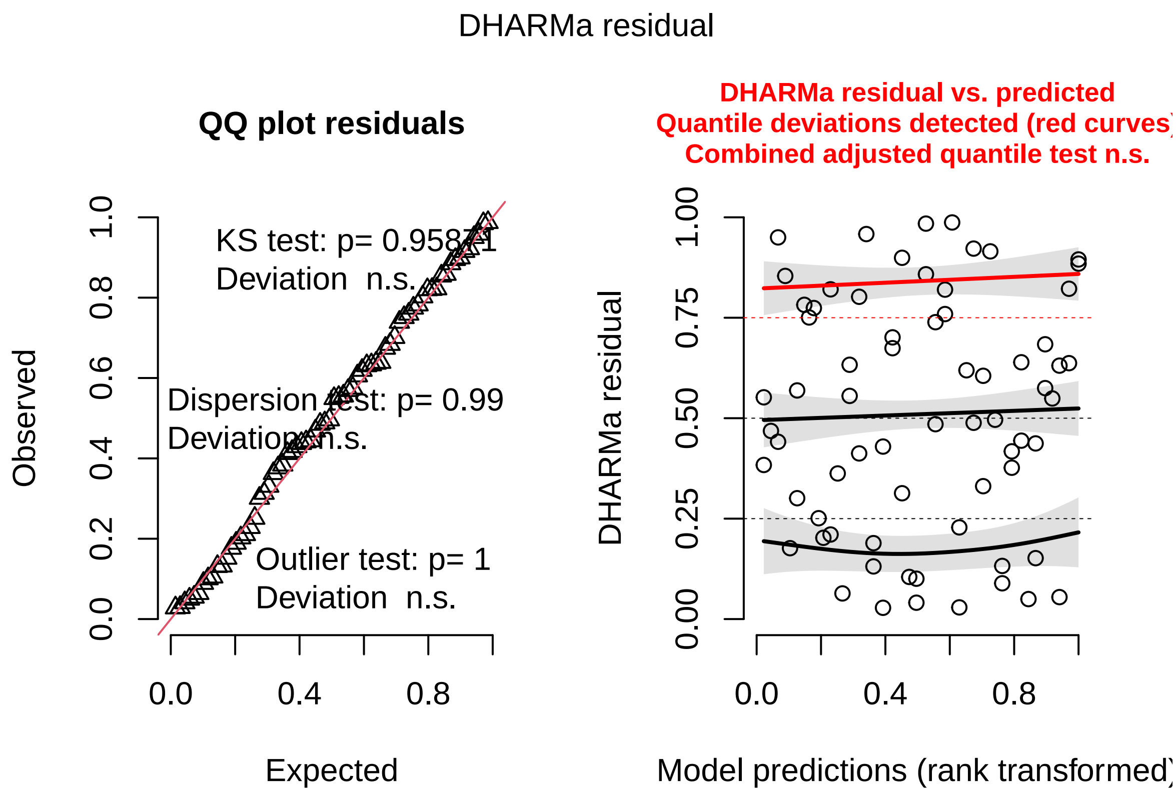
Figure S7 : Distribution of the differences in the performance of the discriminant models trained on true and permuted data. Age status of *F. sanguinea* ants were shuffled in the null variant.

$$\text{head\_width} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + \epsilon,$$

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: head_width ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: -103.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.10138 -0.61076  0.06058  0.43106  2.28980
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.003155 0.05617
##  Residual             0.009110 0.09545
## Number of obs: 68, groups:  colony, 16
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  1.21806    0.02641 38.92702  46.121  < 2e-16 ***
## sang_prop    0.12514    0.04344 62.09975   2.881  0.00544 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.715
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.9861, p-value = 0.6514
```


Model conditional residuals
Frequency
0
5
10
15
20
−0.2
−0.1
0.0
0.1
0.2
Residuals



4.2 Change in the total CHC mass

In this series of models the total CHC mass (normalized by assuming head width of 1.3 mm) was regressed on the proportion of *Formica sanguinea* workers in the colony.

4.2.1 Mature *F. sanguinea*

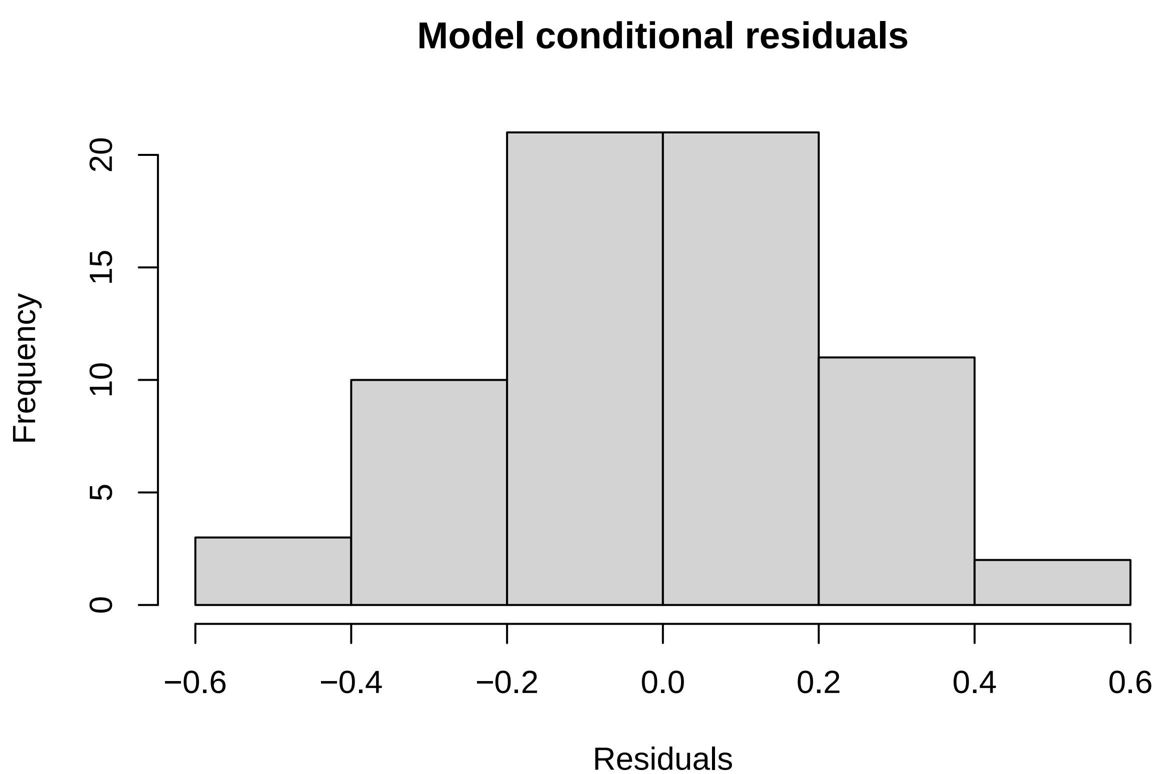
$\log(\text{CHC\_mass\_sanguinea}) = \beta_0 + \text{sanguinea\_proportion} + (\text{head\_width})^2 + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion})$

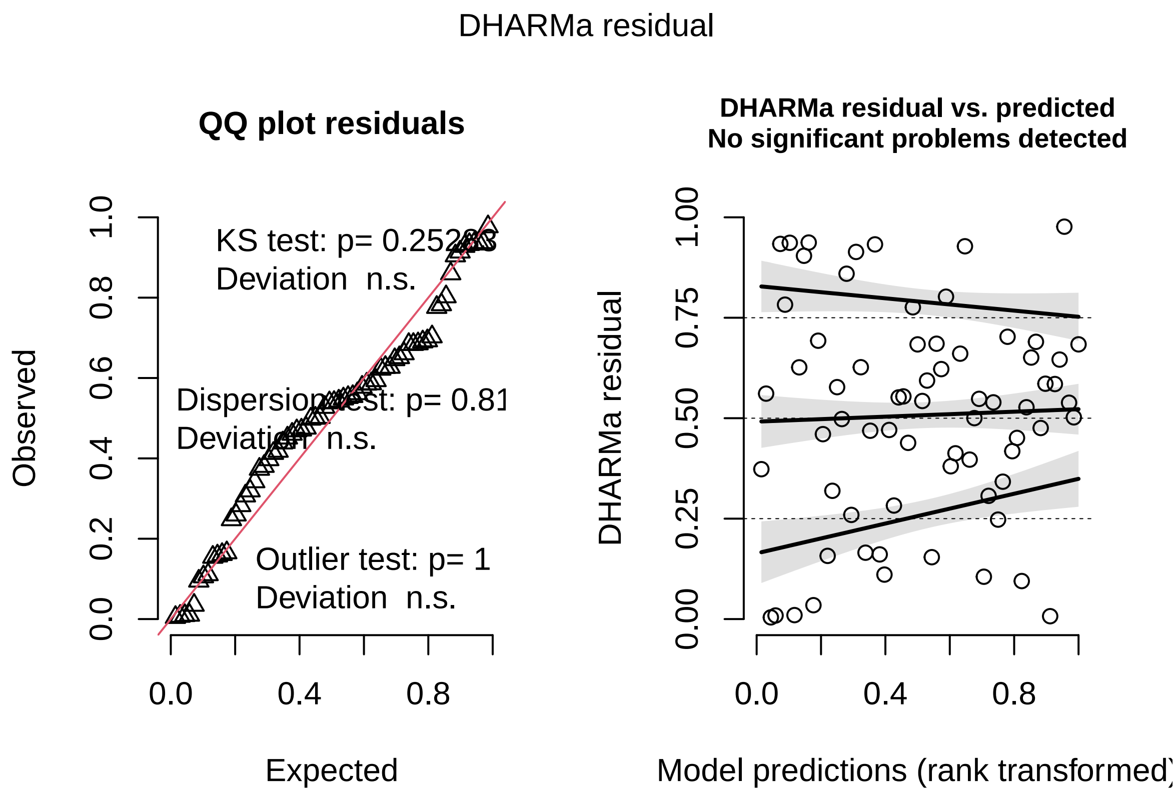
where CHC_mass_sanguinea denotes the total normalized CHC mass on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + I(head_width^2) + (1 | colony) +
##     (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 92.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.71434 -0.36455  0.00409  0.49864  1.78352
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
```

```
##  colony:census_date (Intercept) 0.12525  0.3539
##  colony             (Intercept) 0.03448  0.1857
##  Residual                       0.10396  0.3224
## Number of obs: 68, groups:  colony:census_date, 44; colony, 16
##
## Fixed effects:
##                 Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)       1.2151     0.3429 63.8382   3.544 0.000744 ***
## sang_prop         0.8476     0.2506 39.0054   3.382 0.001646 **
## I(head_width^2)  -0.2012     0.2134 61.6709  -0.943 0.349440
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) sng_pr
## sang_prop    0.000
## I(hd_wdt^2) -0.924 -0.305
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98867, p-value = 0.798
```





The model shows no significant effect of body size on CHC amount in *F. sanguinea*, and this term will therefore be excluded from subsequent models for this species.

4.2.2 *F. fusca*

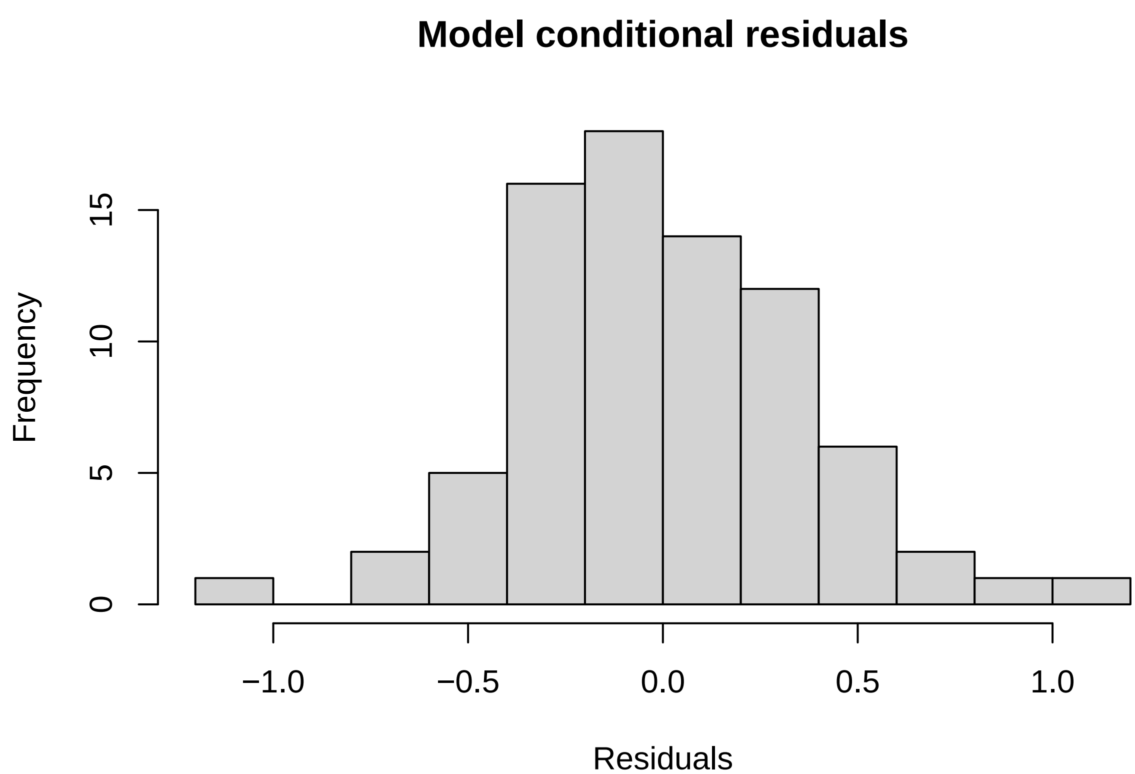
$$\log(\text{CHC\_mass\_fusca}) = \beta_0 + \text{sanguinea\_proportion} + (\text{head\_width})^2 + (1|\text{colony\_ID}) + \epsilon,$$

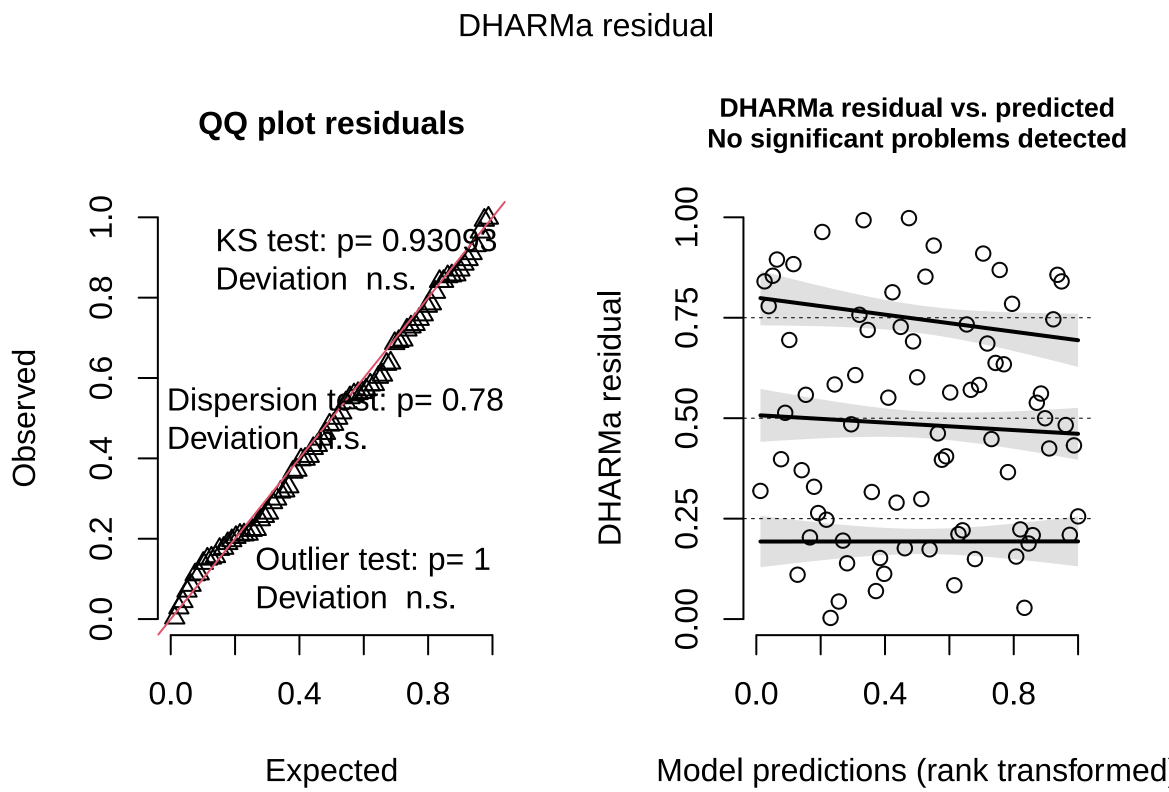
where CHC_mass_fusca denotes the total normalized CHC mass on the body of adult *F. fusca* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + I(head_width^2) + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 99.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7162 -0.5731 -0.1218  0.5782  2.4978
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.05605  0.2367
##  Residual             0.16466  0.4058
## Number of obs: 78, groups:  colony, 20
##
```

```
## Fixed effects:
##                  Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)        1.5118     0.3851 58.0198   3.926 0.000232 ***
## sang_prop          0.2627     0.1738 68.8327   1.511 0.135262
## I(head_width^2)   -0.8355     0.3270 59.7153  -2.555 0.013192 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) sng_pr
## sang_prop     0.077
## I(hd_wdt^2) -0.974 -0.210
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99146, p-value = 0.8874
```





4.2.3 Callow *F. sanguinea*

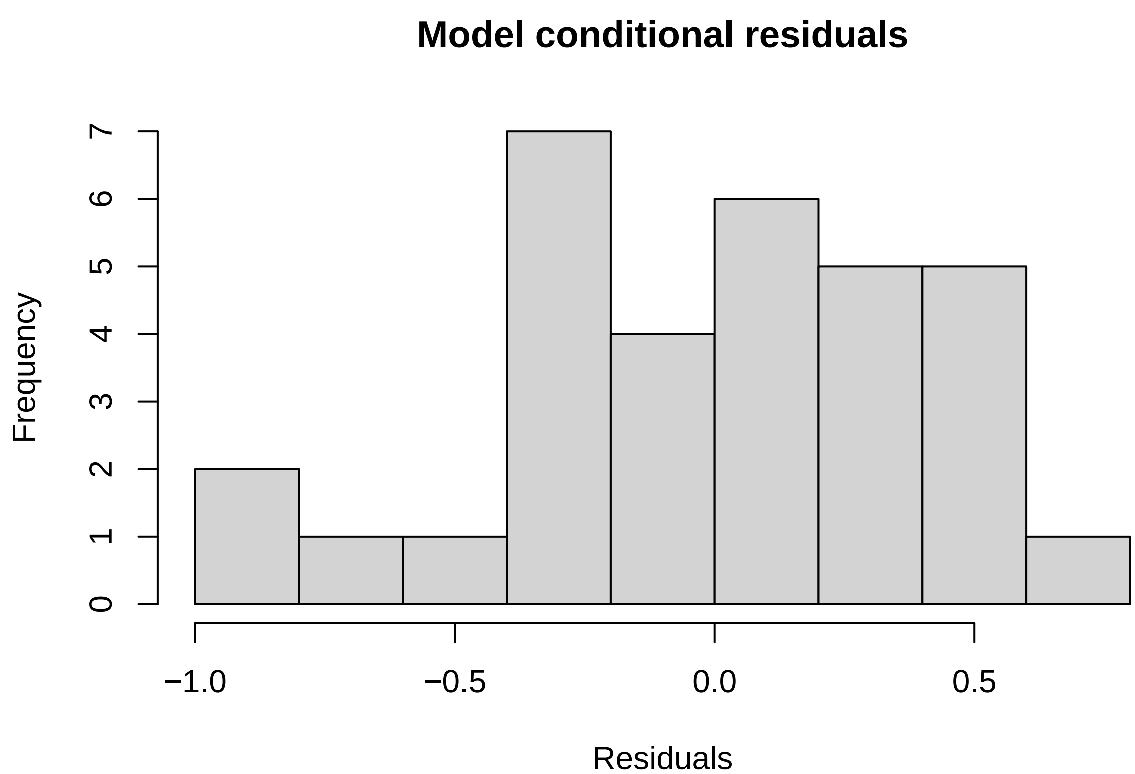
$\log(\text{CHC\_mass\_callow}) = \beta_0 + \text{sanguinea\_proportion} + (\text{head\_width})^2 + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_II}$

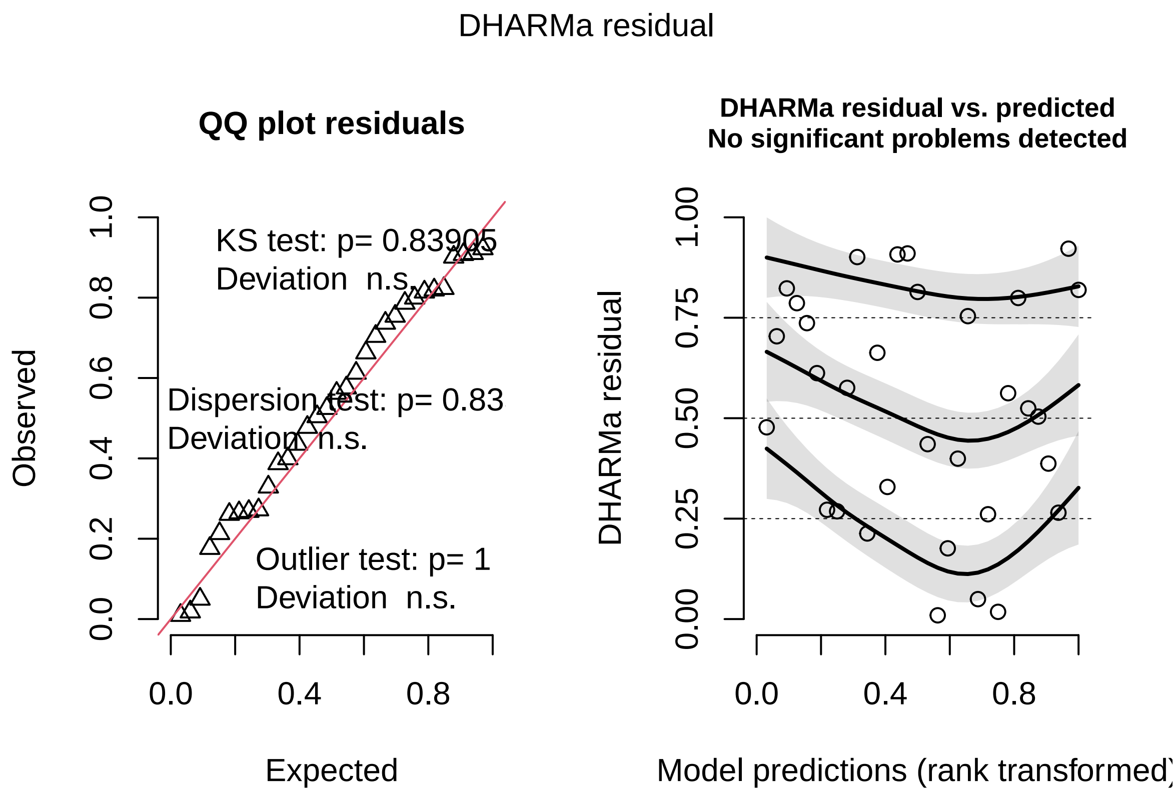
where CHC_mass_callow denotes the total normalized CHC mass on the body of callow *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + I(head_width^2) + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 77.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.94324 -0.51856  0.06638  0.66444  1.47139
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.07045  0.2654
##  Residual             0.21112  0.4595
## Number of obs: 32, groups:  colony, 16
##
## Fixed effects:
##                     Estimate Std. Error         df t value Pr(>|t|)
```

```
## (Intercept)      5.190e-01  5.136e-01  2.893e+01   1.010  0.32071
## sang_prop        1.042e+00  2.968e-01  2.431e+01   3.510  0.00177 **
## I(head_width^2) -4.705e-07  2.899e-07  2.808e+01  -1.623  0.11576
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) sng_pr
## sang_prop   -0.295
## I(hd_wdt^2) -0.945  0.047
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.95837, p-value = 0.2478
```





Similarly to mature *F. sanguinea* ants, in callow workers there is negative but not significant relationship between body size and normalized CHC amount.

4.3 Change of the CHC characteristic of *F. sanguinea*

In this section the subset of peaks were used to calculate normalized CHC mass. These were peaks identified as *F. sanguinea* markers.

4.3.1 Mature *F. sanguinea* ants

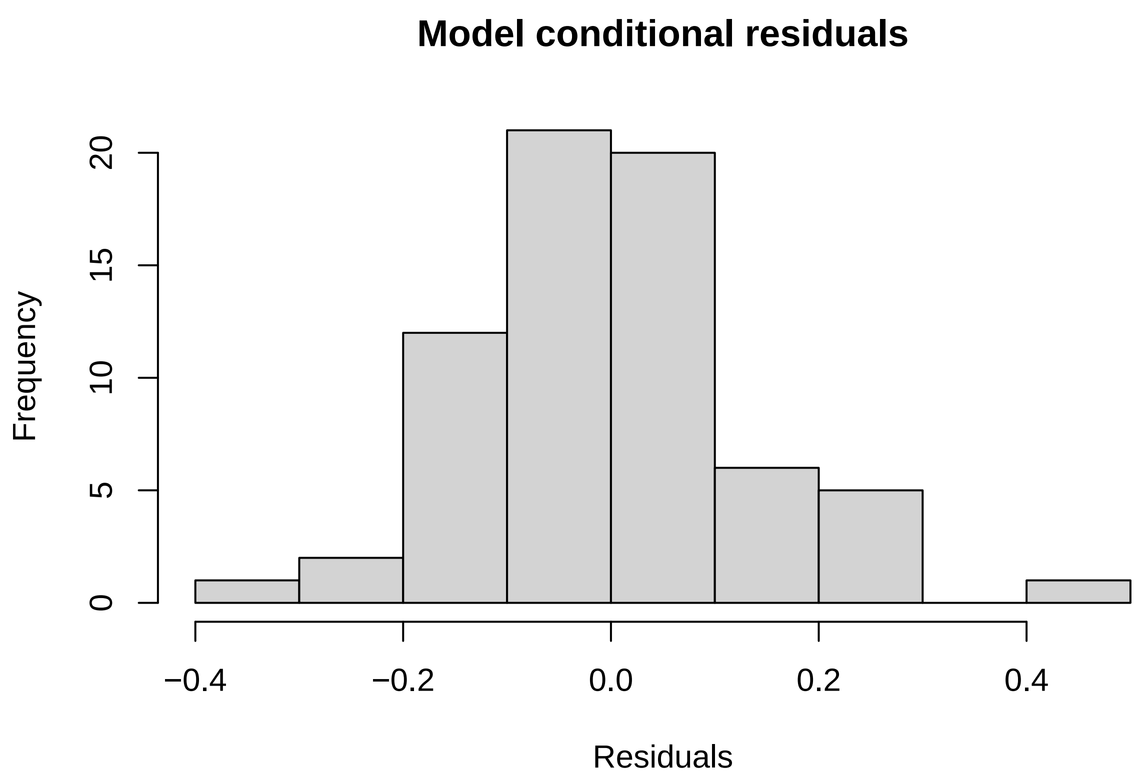
$$\log(\text{CHC\_mass\_mature}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

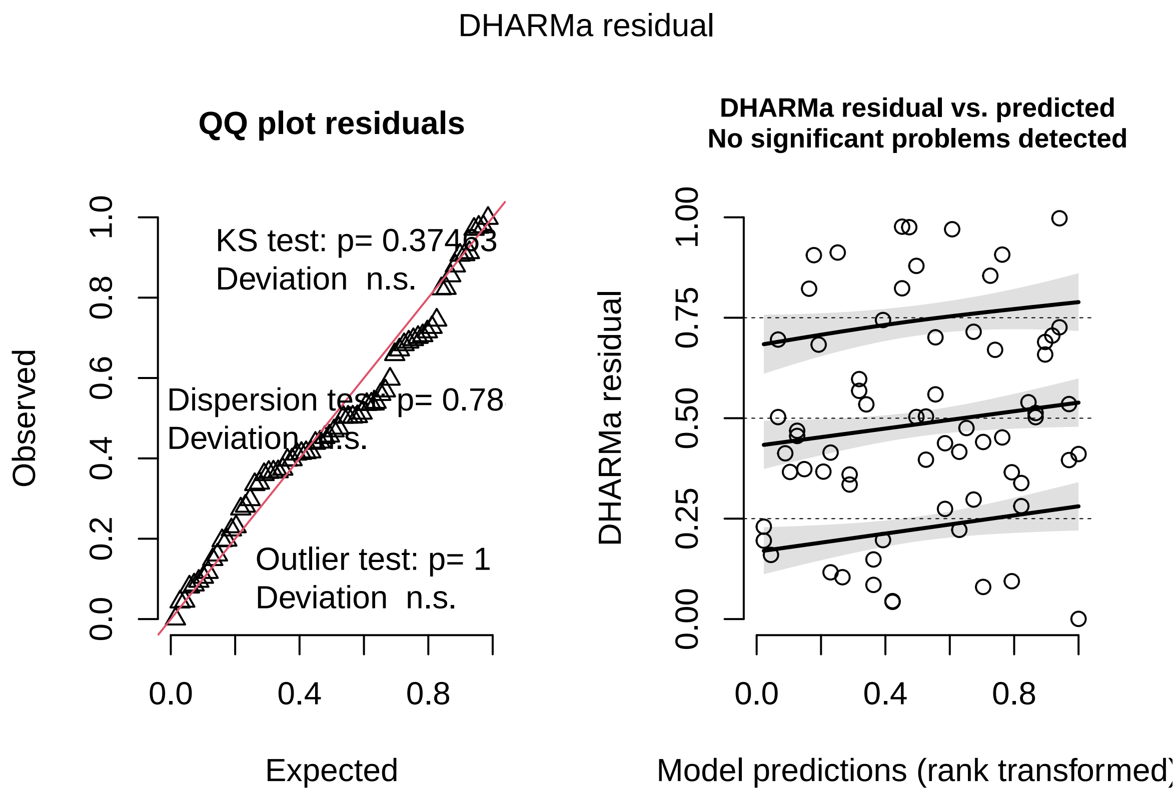
where CHC_mass_mature denotes the normalized mass of *F. sanguinea* markers on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass + 1)) ~ sang_prop + (1 | colony:census_date) +      (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 12.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.97298 -0.34447 -0.03782  0.36484  2.33544
```

```
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.04798  0.2190
##  colony             (Intercept) 0.00000  0.0000
##  Residual                       0.02983  0.1727
## Number of obs: 68, groups:  colony:census_date, 44; colony, 16
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  0.21163    0.06959 40.15928   3.041  0.00414 **
## sang_prop    1.13967    0.13521 39.59851   8.429 2.27e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.823
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98236, p-value = 0.4494
```





4.3.2 *F. fusca* ants

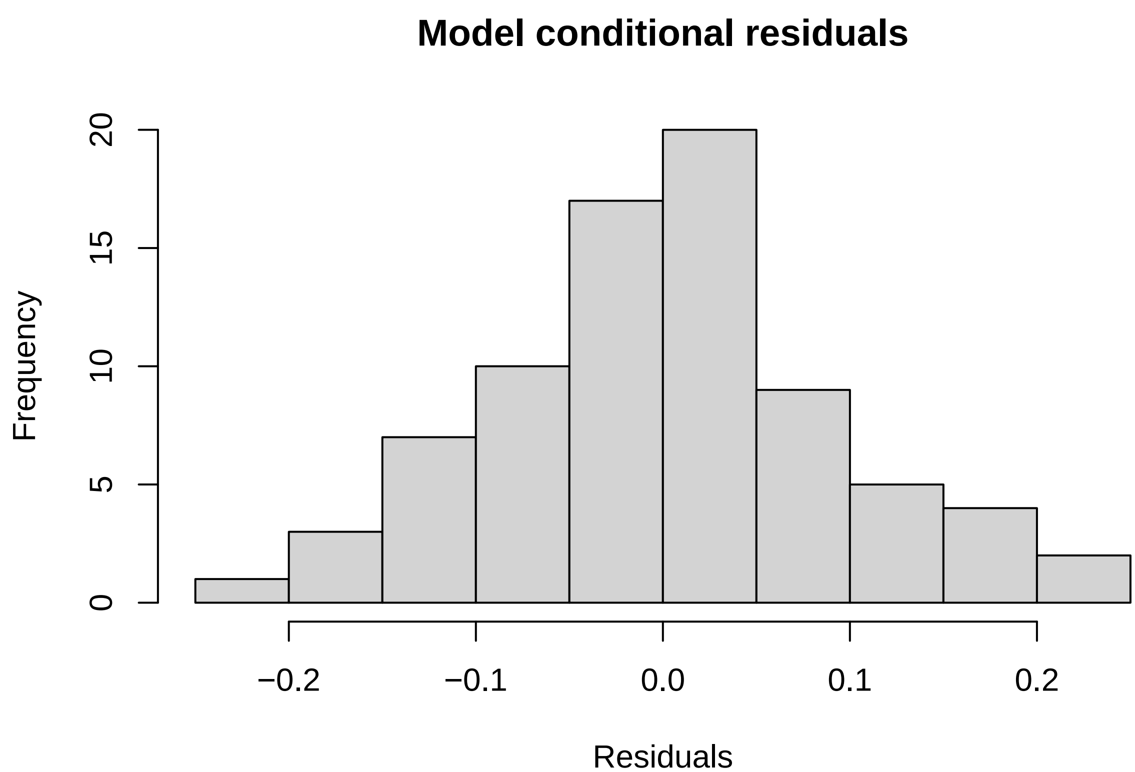
$\sqrt{(\text{CHC\_mass\_fusca})} = \beta_0 + \text{sanguinea\_proportion} + \text{head\_width}^2 + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) +$

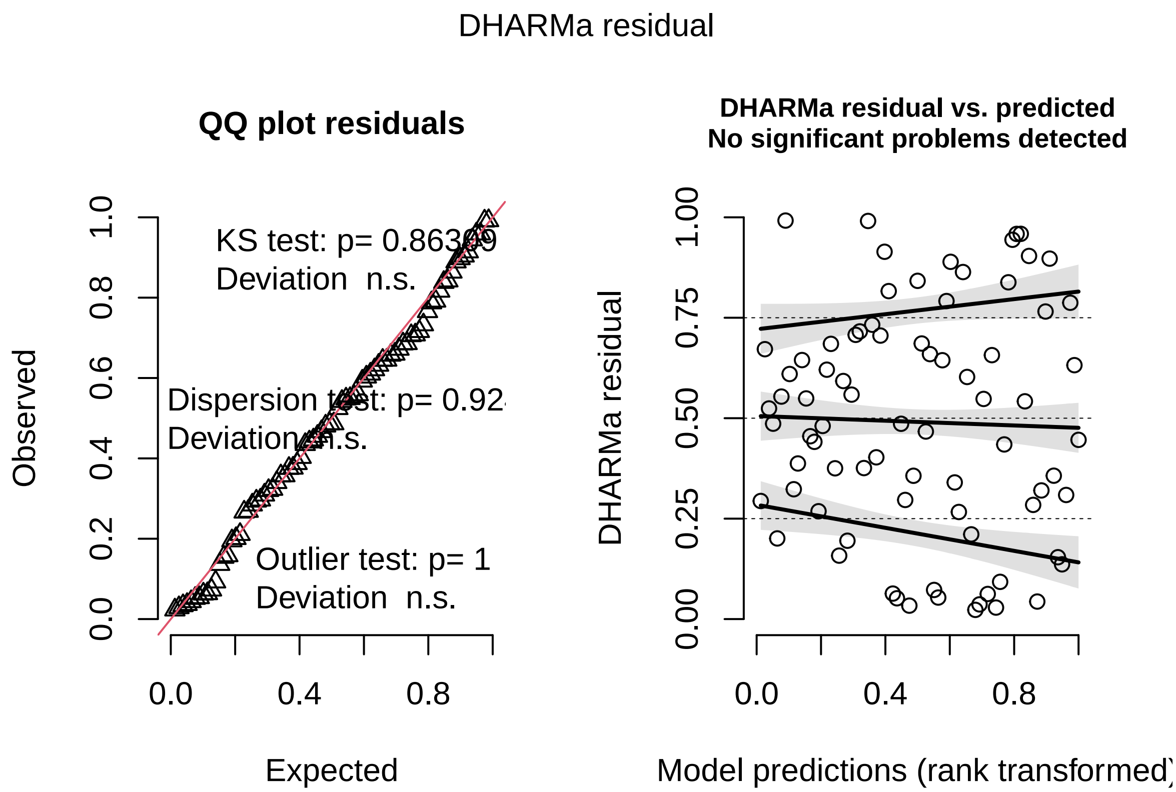
where CHC_mass_fusca denotes the normalized mass of *F. sanguinea* markers on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(corrected_mass)) ~ sang_prop + I(head_width^2) + (1 |      colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: -77.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.01554 -0.48607  0.05576  0.43926  2.03376
##
## Random effects:
##  Groups              Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.007315 0.08553
##  Residual                        0.012634 0.11240
## Number of obs: 78, groups:  colony:census_date, 55
##
## Fixed effects:
##                 Estimate Std. Error       df t value Pr(>|t|)
```

```
## (Intercept)       0.53201    0.11050 66.92660   4.815 8.78e-06 ***
## sang_prop         0.55826    0.06183 51.54390   9.029 3.39e-12 ***
## I(head_width^2) -0.15001    0.09372 67.96260  -1.601    0.114
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) sng_pr
## sang_prop    0.054
## I(hd_wdt^2) -0.973 -0.224
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99034, p-value = 0.8276
```





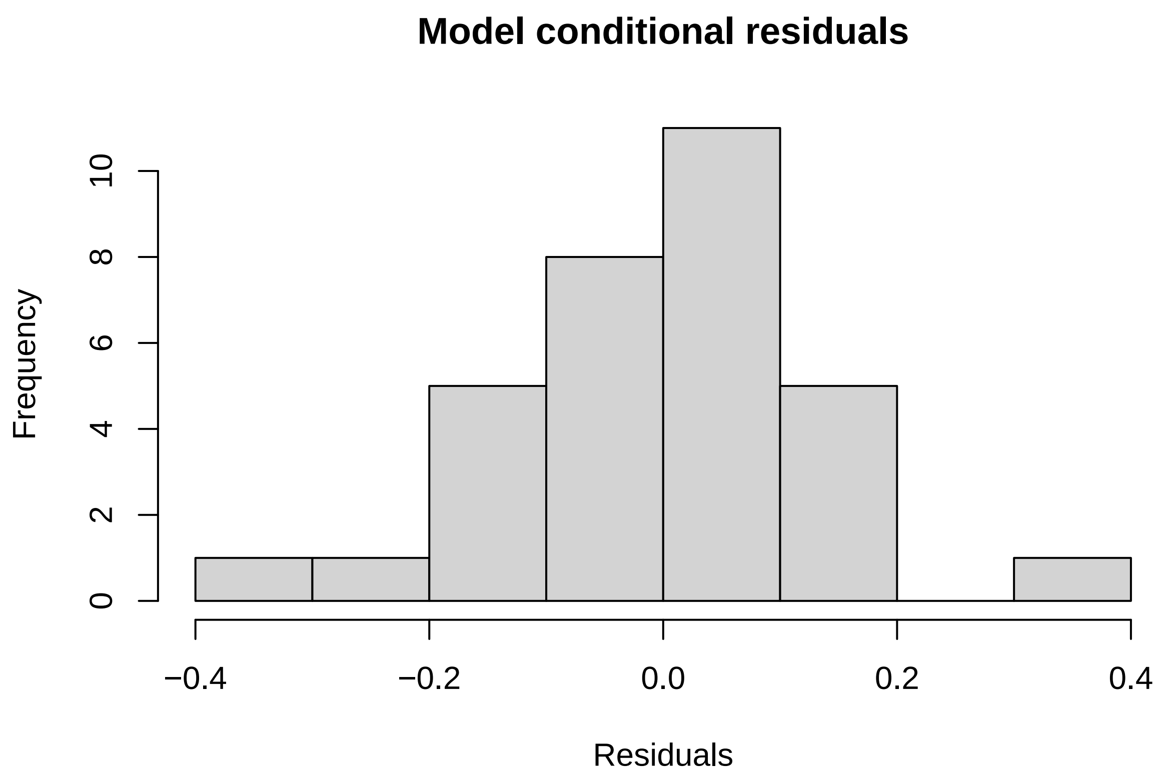
4.3.3 Callow *F. sanguinea* ants

$$\sqrt[3]{(\text{CHC\_mass\_mature})} = \beta_0 + \text{sanguinea\_proportion} + \epsilon,$$

where CHC_mass_mature denotes the normalized mass of *F. sanguinea* markers on the body of callow *F. sanguinea* worker.

```
##
## Call:
## lm(formula = I((corrected_mass)^(1/3)) ~ sang_prop, data = model_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37797 -0.06316  0.02260  0.07389  0.32290
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.28572    0.04235   6.746 1.77e-07 ***
## sang_prop    0.65147    0.07962   8.182 3.91e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1345 on 30 degrees of freedom
## Multiple R-squared:  0.6906, Adjusted R-squared:  0.6803
## F-statistic: 66.95 on 1 and 30 DF,  p-value: 3.915e-09
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.96876, p-value = 0.4658
```



4.4 Change of the CHC characteristic of *F. fusca*

Similarly to the analysis with the use of markers of *F. sanguinea* workers, the procedure was repeated using the mass of *F. fusca* markers as a response variable.

4.4.1 Mature *F. sanguinea* ants

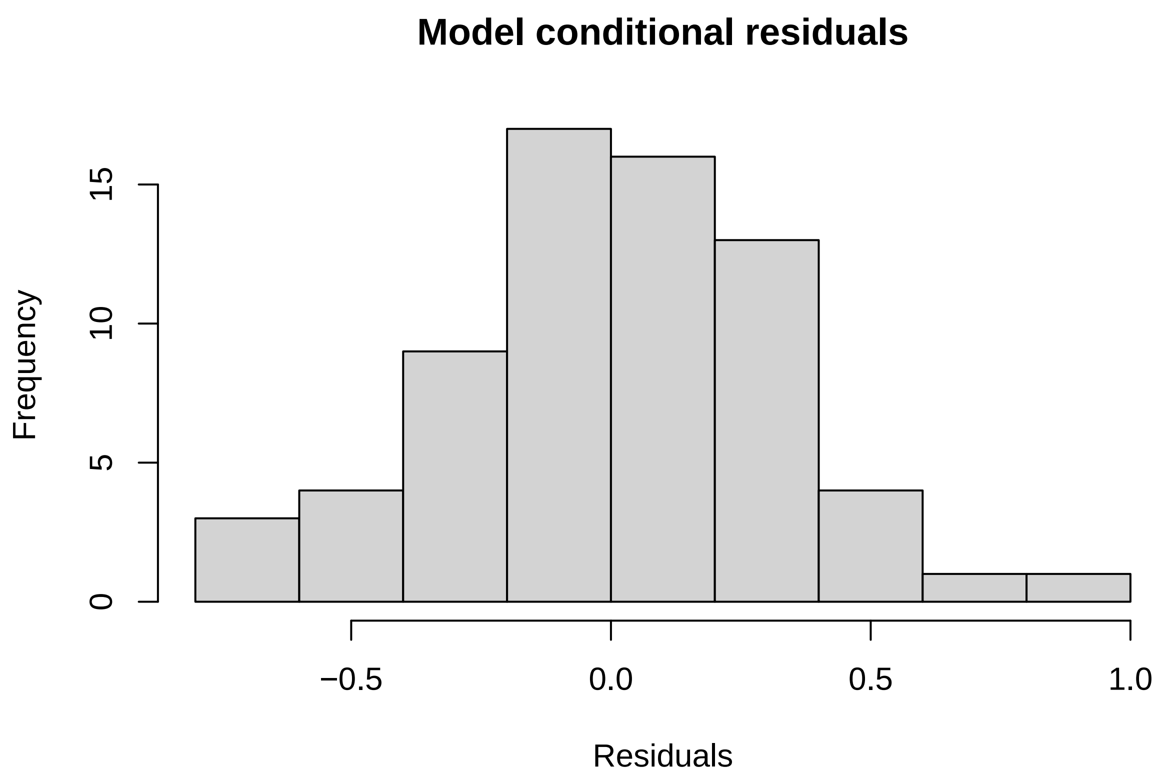
$$\log(\text{CHC\_mass\_mature}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

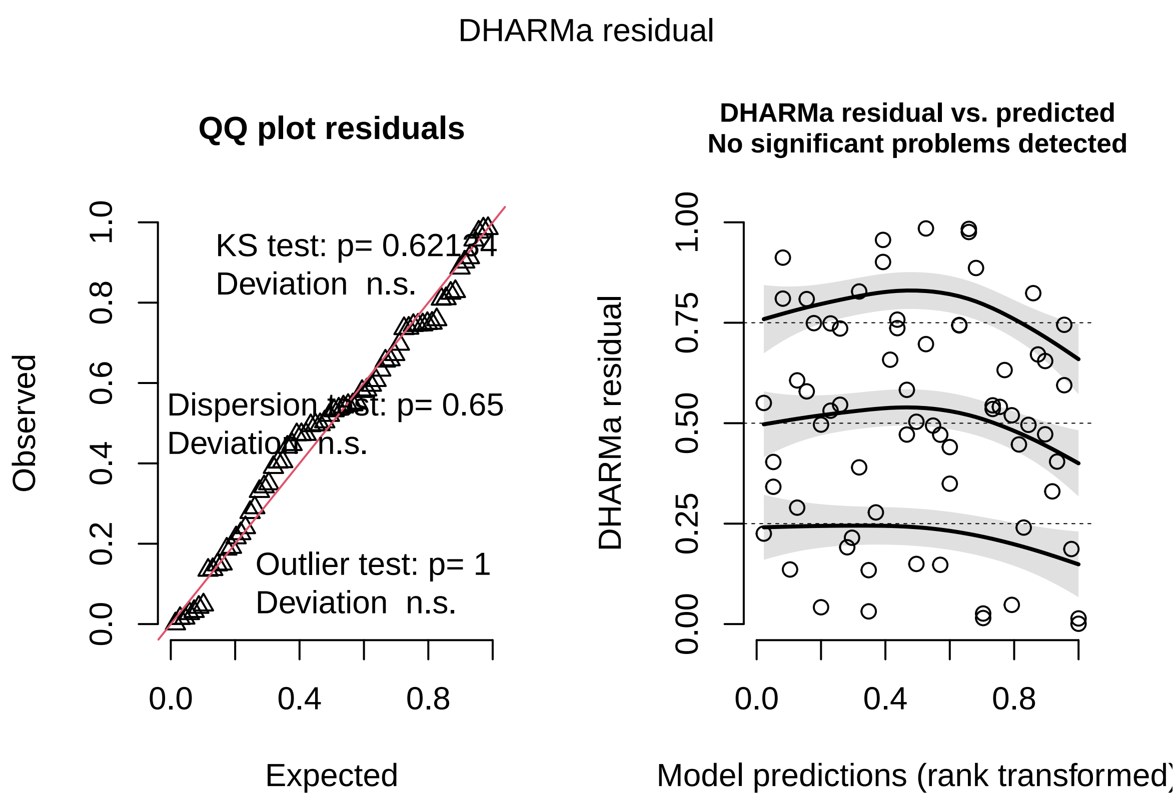
where CHC_mass_mature denotes the normalized mass of *F. fusca* markers on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + (1 | colony:census_date) +      (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 132
##
```

```
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.82132 -0.46113  0.04855  0.54966  1.99234
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.1494   0.3866
##  colony             (Intercept) 0.2376   0.4875
##  Residual                       0.1826   0.4273
## Number of obs: 68, groups:  colony:census_date, 44; colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.7444     0.1958 33.8649  -3.802 0.000571 ***
## sang_prop    -1.1209     0.2971 33.3261  -3.772 0.000632 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.657
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98893, p-value = 0.8114
```


Model conditional residuals
Frequency
0
5
10
15
−0.5
0.0
0.5
1.0
Residuals



4.4.2 *F. fusca* ants

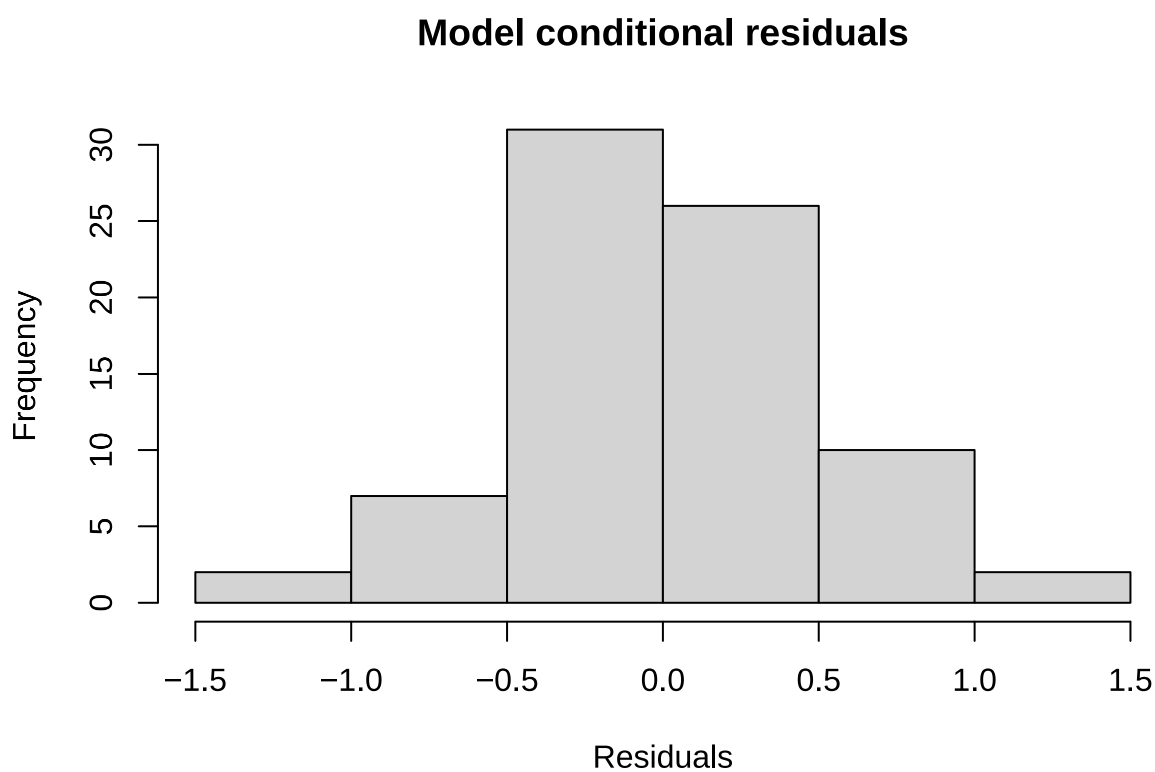
$\log(\text{CHC\_mass\_mature}+0.01) = \beta_0 + \text{sanguinea\_proportion} + \text{head\_width}^2 + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasio}$

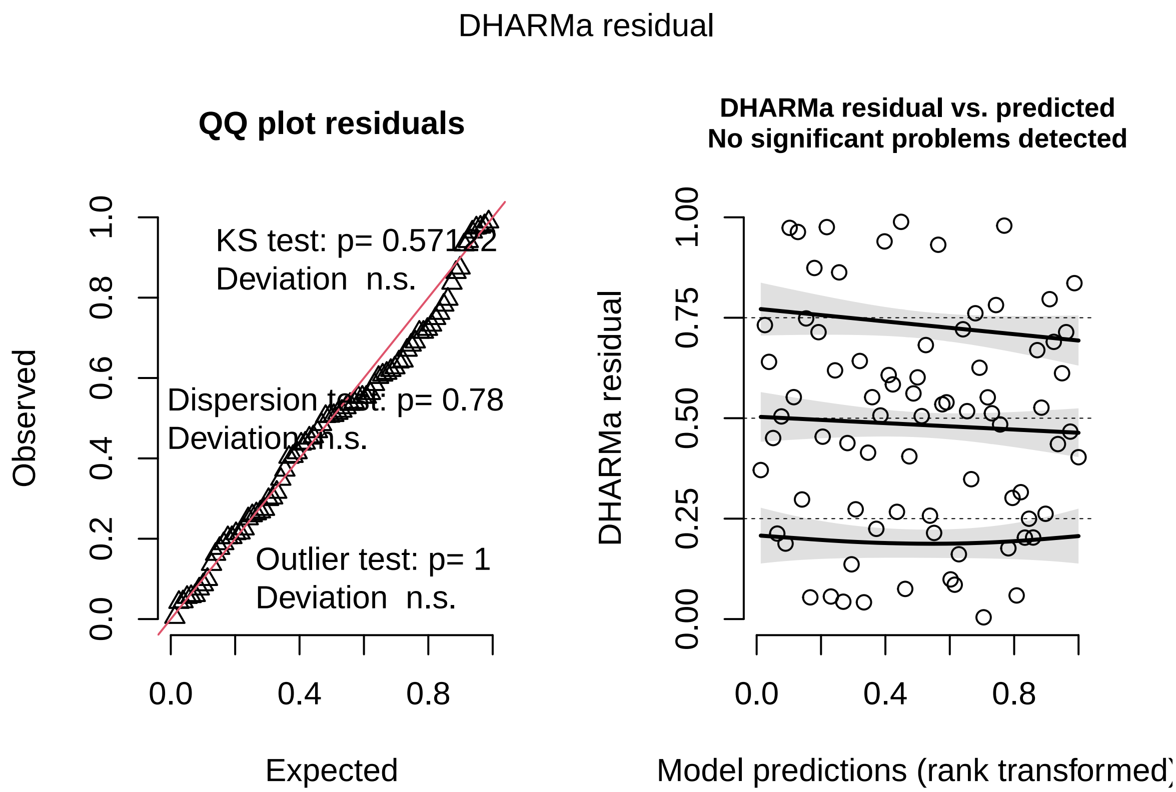
where CHC_mass_mature denotes the normalized mass of *F. fusca* markers on the body of *F. fusca* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass + 0.01)) ~ sang_prop + I(head_width^2) +
##     (1 | colony) + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 155.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.25861 -0.56138 -0.00894  0.52028  2.15785
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.0000   0.0000
##  colony             (Intercept) 0.2471   0.4971
##  Residual                       0.3042   0.5515
## Number of obs: 78, groups:  colony:census_date, 55; colony, 20
##
## Fixed effects:
```

```
##                  Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)       -0.2641     0.5811 73.2247  -0.454   0.6508
## sang_prop         -0.7400     0.2434 64.8240  -3.041   0.0034 **
## I(head_width^2)  -0.7359     0.4904 74.5636  -1.500   0.1377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) sng_pr
## sang_prop     0.091
## I(hd_wdt^2) -0.967 -0.212
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99125, p-value = 0.8772
```





4.4.3 Callow *F. sanguinea* ants

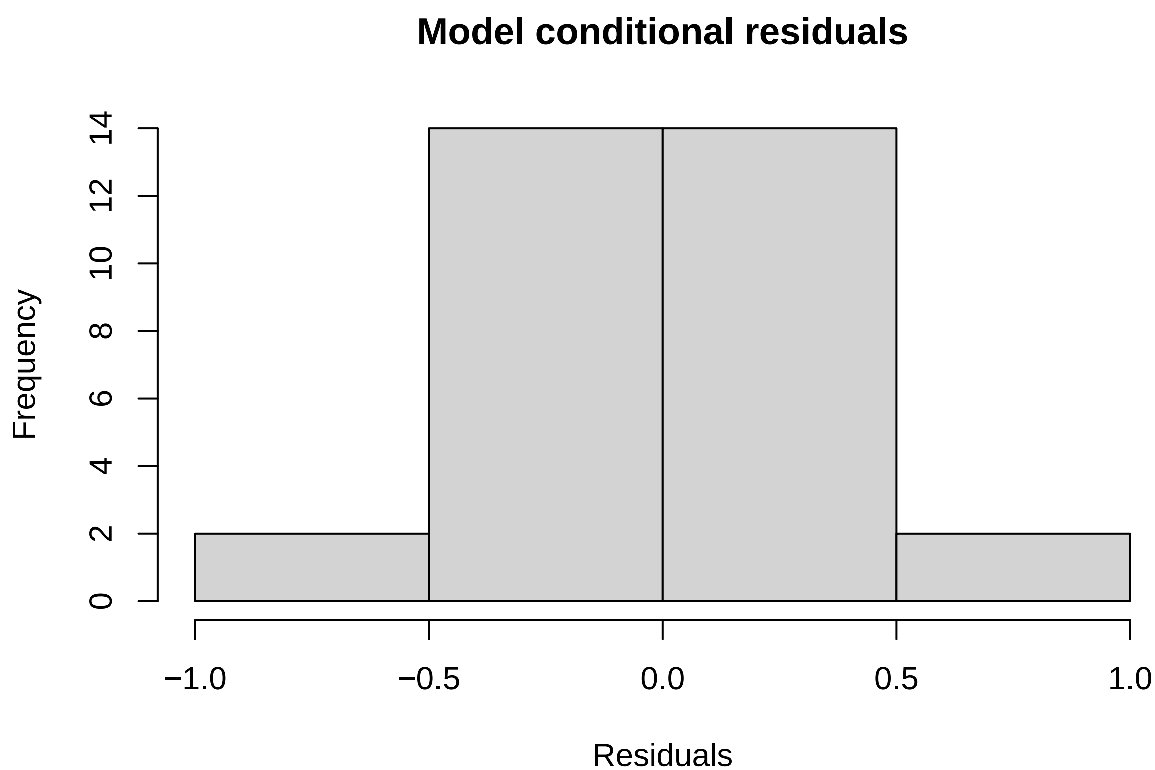
$$\log(\text{CHC\_mass\_mature}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + \epsilon,$$

where log(CHC_mass_mature) denotes the normalized mass of *F. fusca* markers on the body of callow *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 60
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.91742 -0.57905  0.00403  0.67341  1.87987
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.2132   0.4617
##  Residual             0.2274   0.4769
## Number of obs: 32, groups:  colony, 16
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
```

```
## (Intercept) -1.96642    0.20293 28.17254  -9.690  1.8e-10 ***
## sang_prop   -0.07448    0.32555 19.98435  -0.229    0.821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.691
```

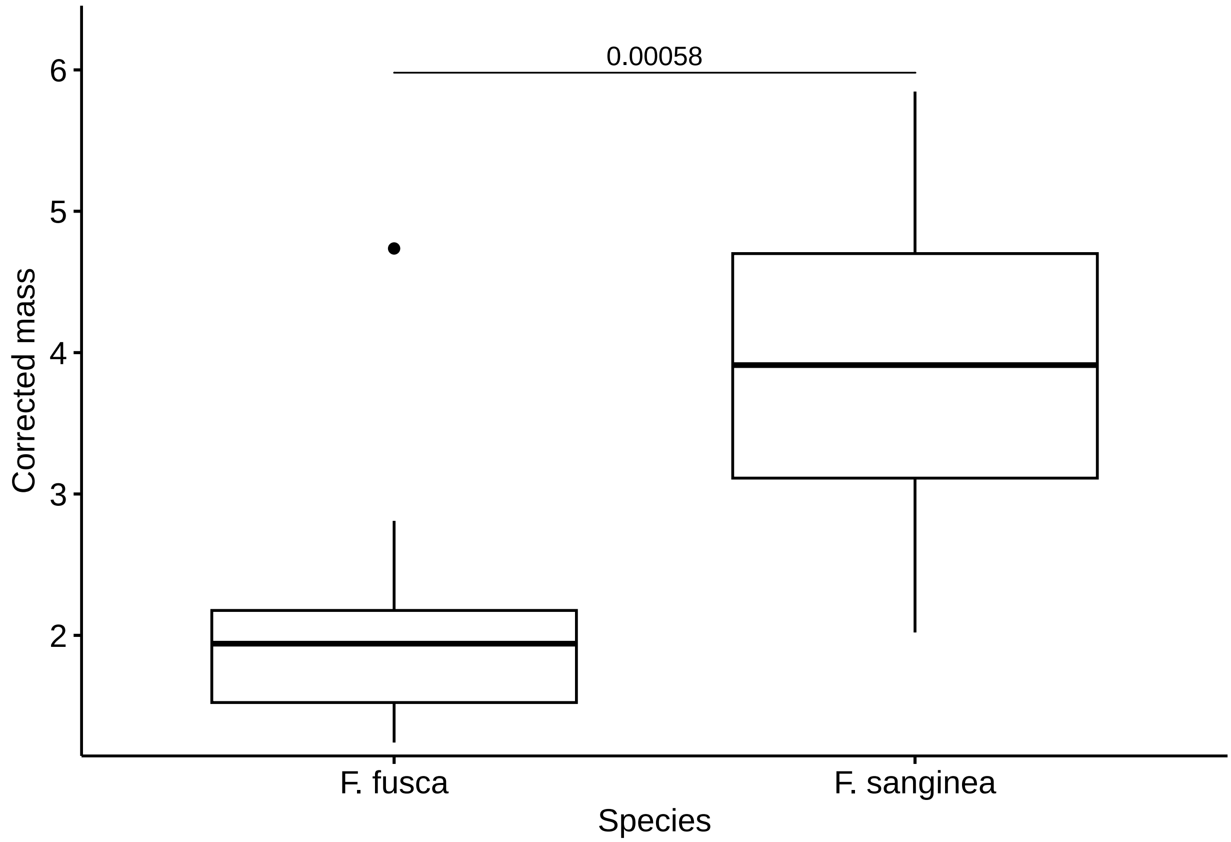
```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98685, p-value = 0.9565
```



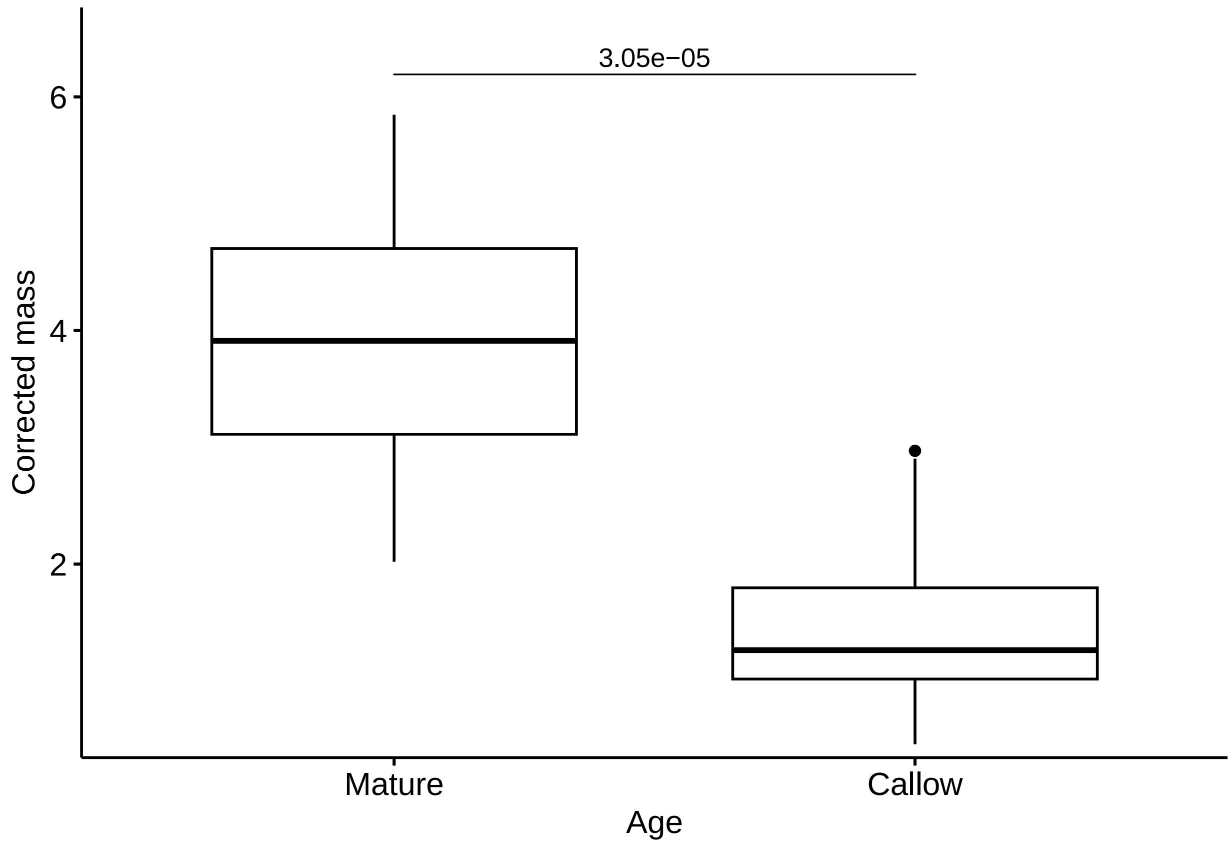
5 Comparison of the amount and proportions of CHC between species and age categories

This section presents the results of Wilcoxon tests comparing features of CHC profiles between callow and mature *F. sanguinea* ants as well as *F. fusca* slaves. Samples from the same colony were averaged before calculation of the final statistics to account for their non-independence.

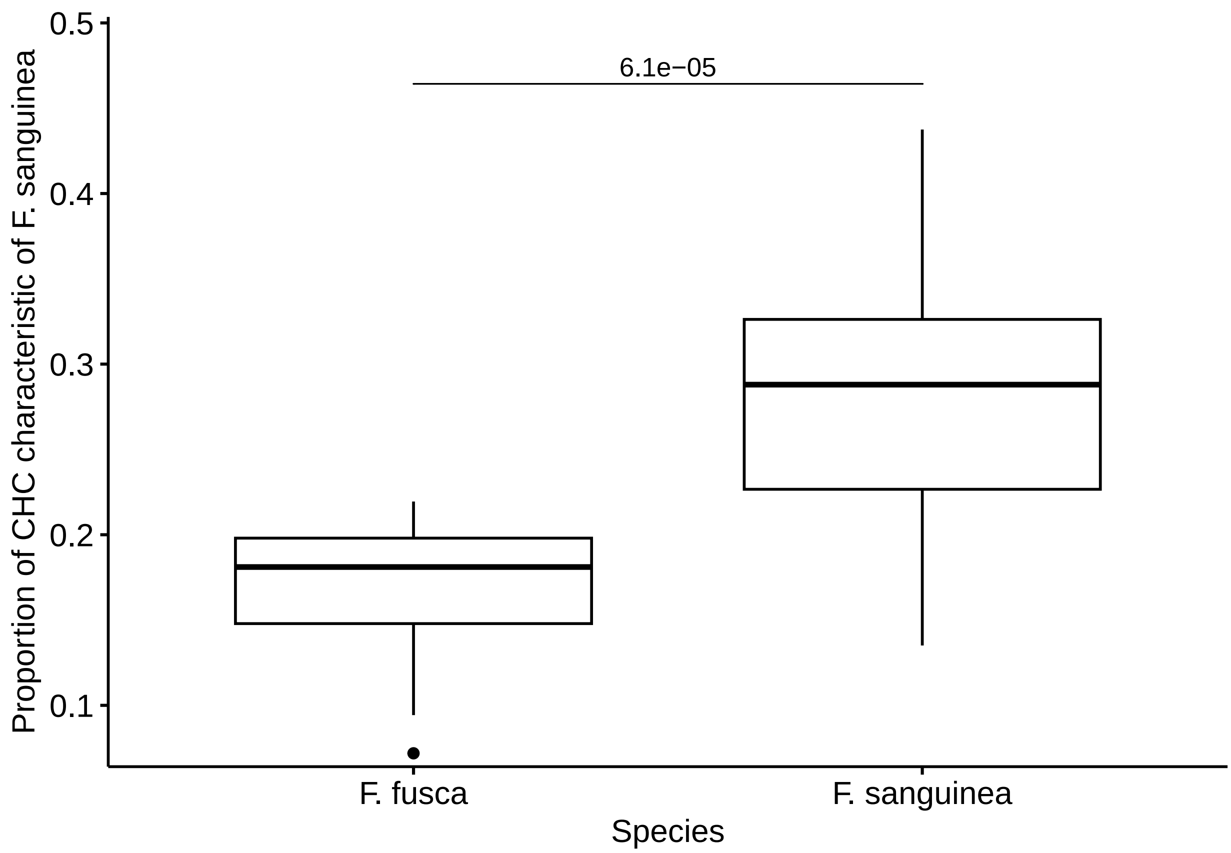
5.1 Difference in the total CHC amount between *F. sanguinea* and *F. fusca*



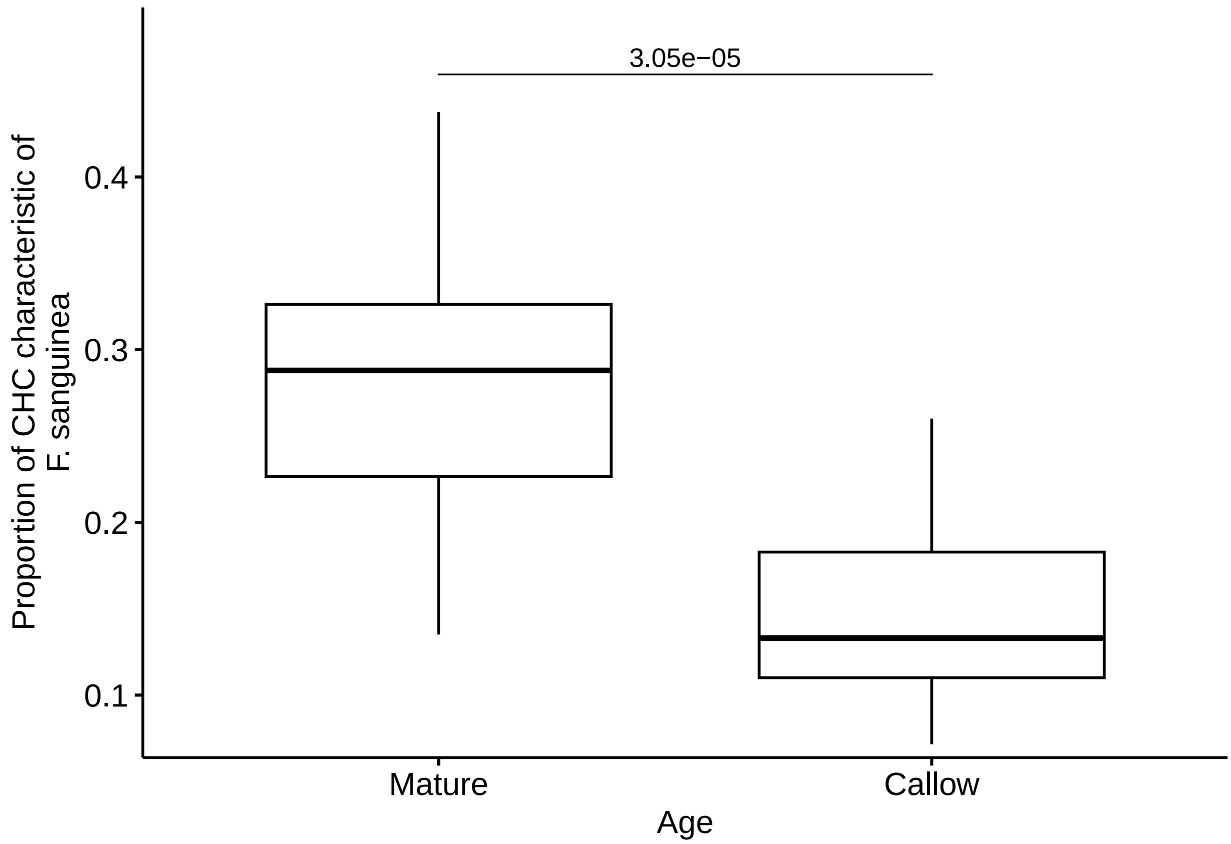
5.2 Difference in the total CHC amount between callow and mature *F. sanguinea*



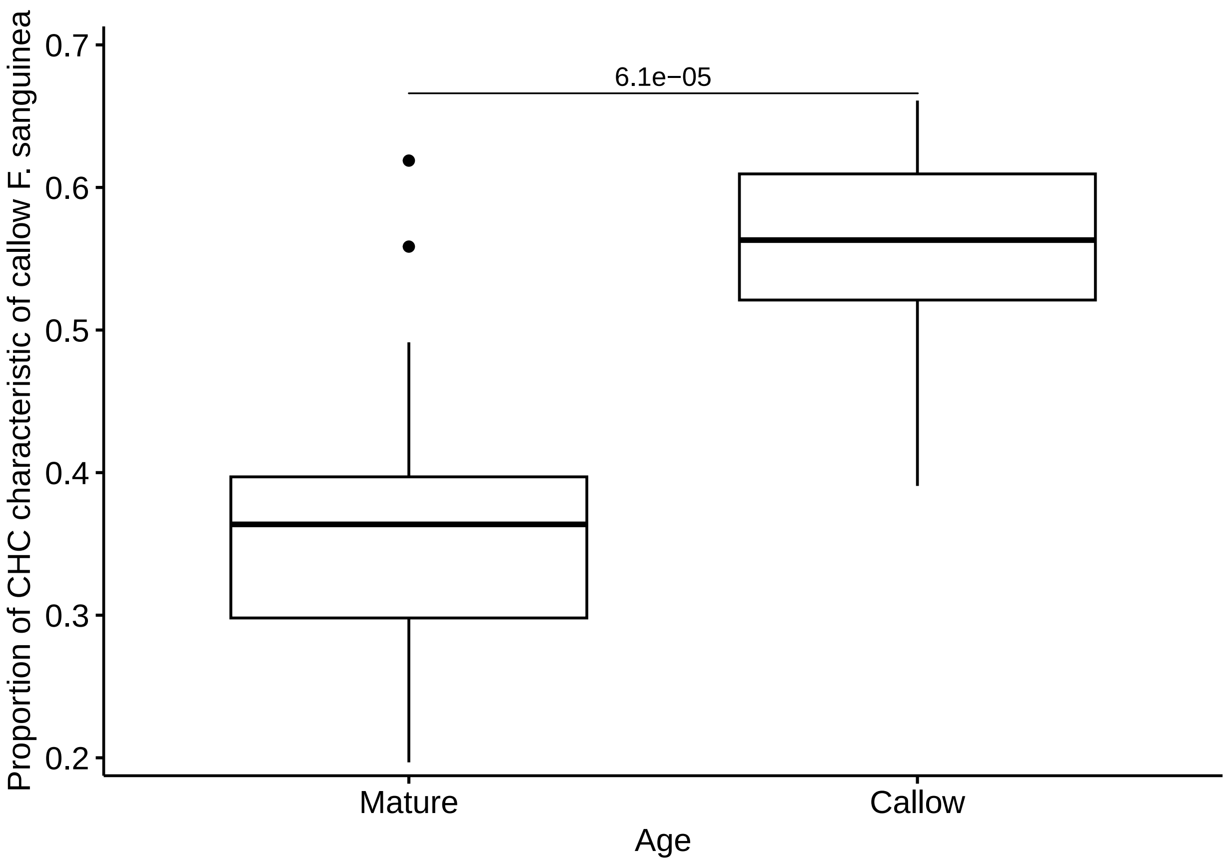
5.3 Difference in the proportion of CHC characteristic of *F. sanguinea* between mature *F. sanguinea* and *F. fusca*



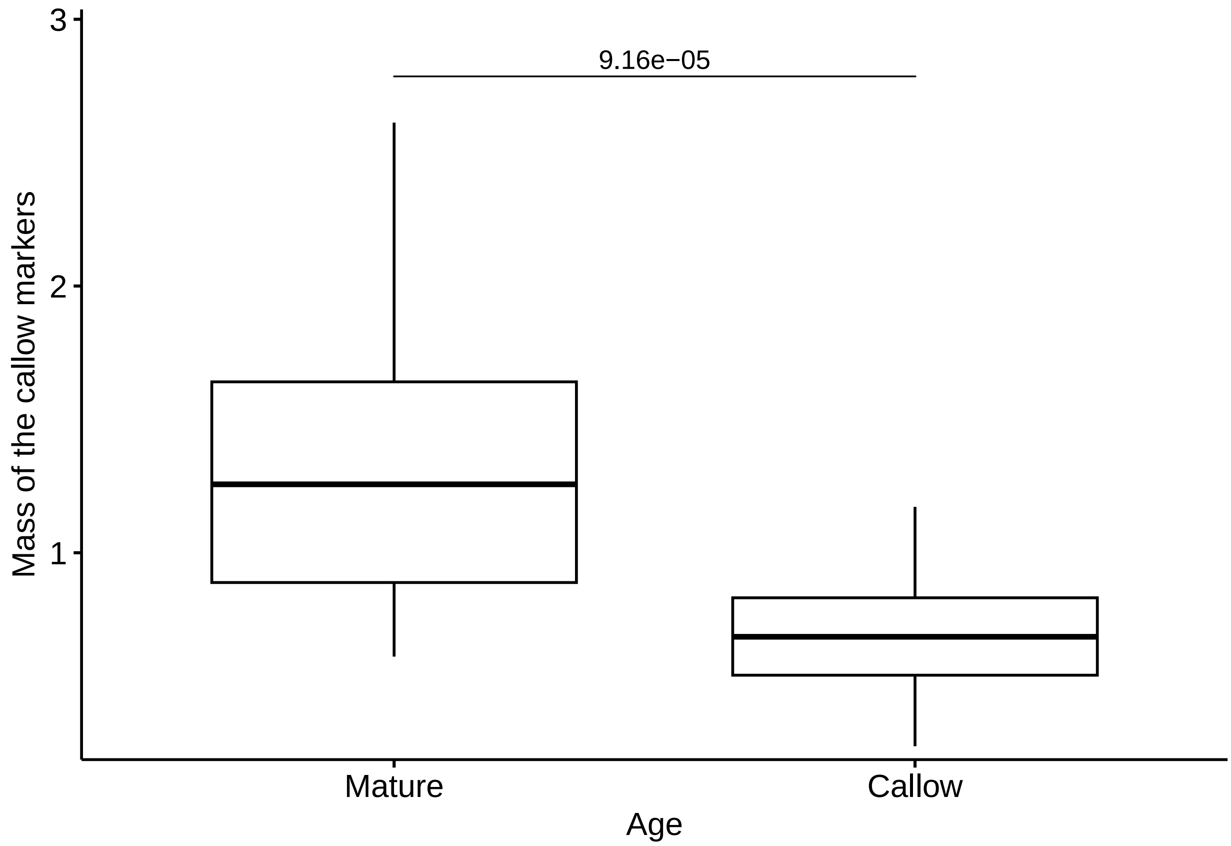
5.4 Difference in the proportion of CHC characteristic of *F. sanguinea* between mature and callow *F. sanguinea*



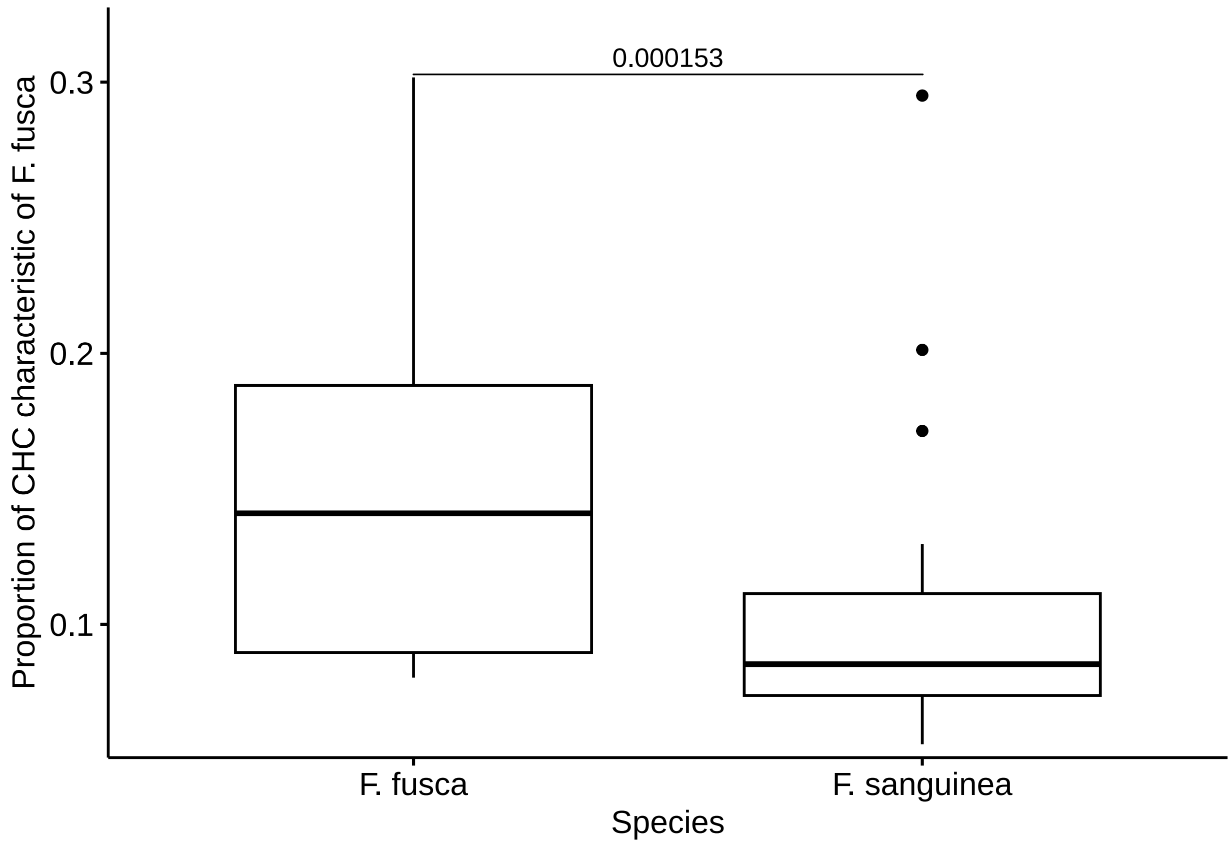
5.5 Difference in the proportion of CHC characteristic of callow *F. sanguinea* between mature and callow *F. sanguinea*



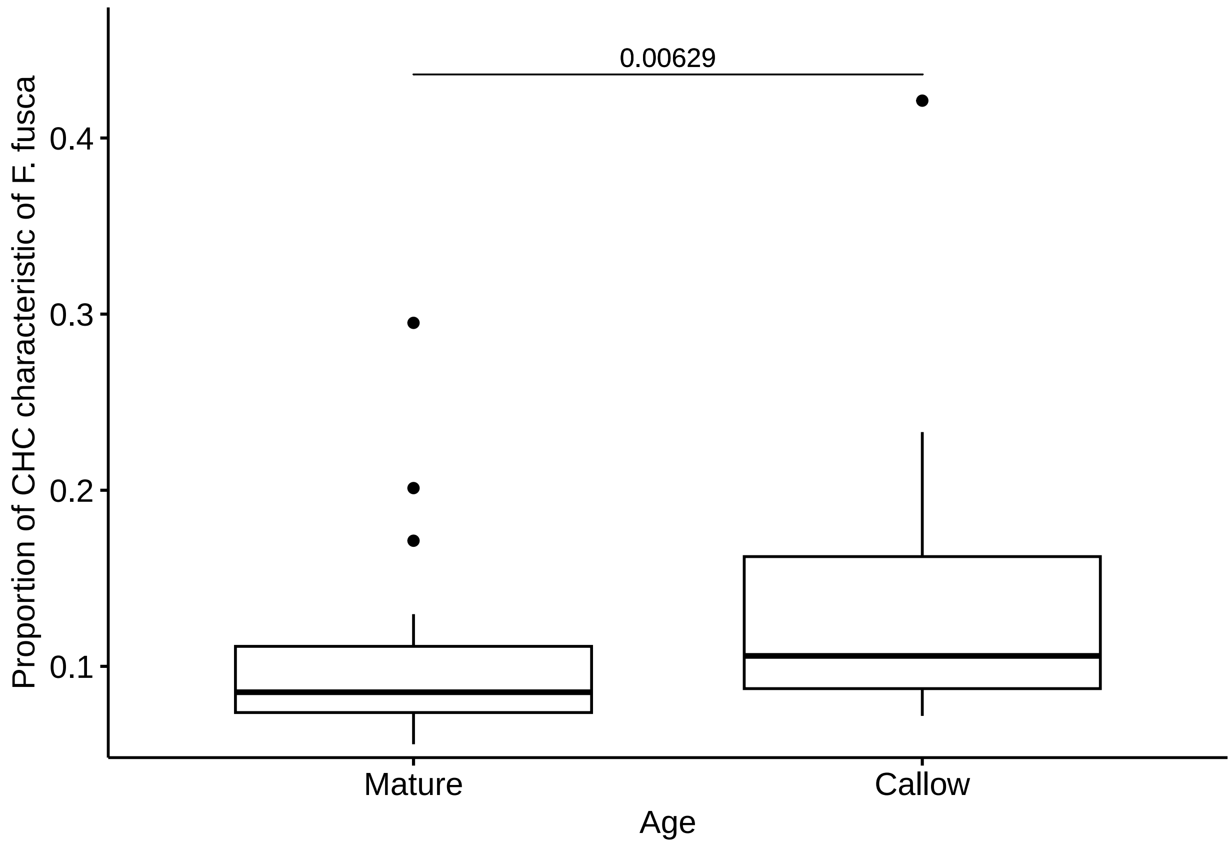
5.6 Difference in the amount of CHC characteristic of callow *F. sanguinea* between mature and callow *F. sanguinea*



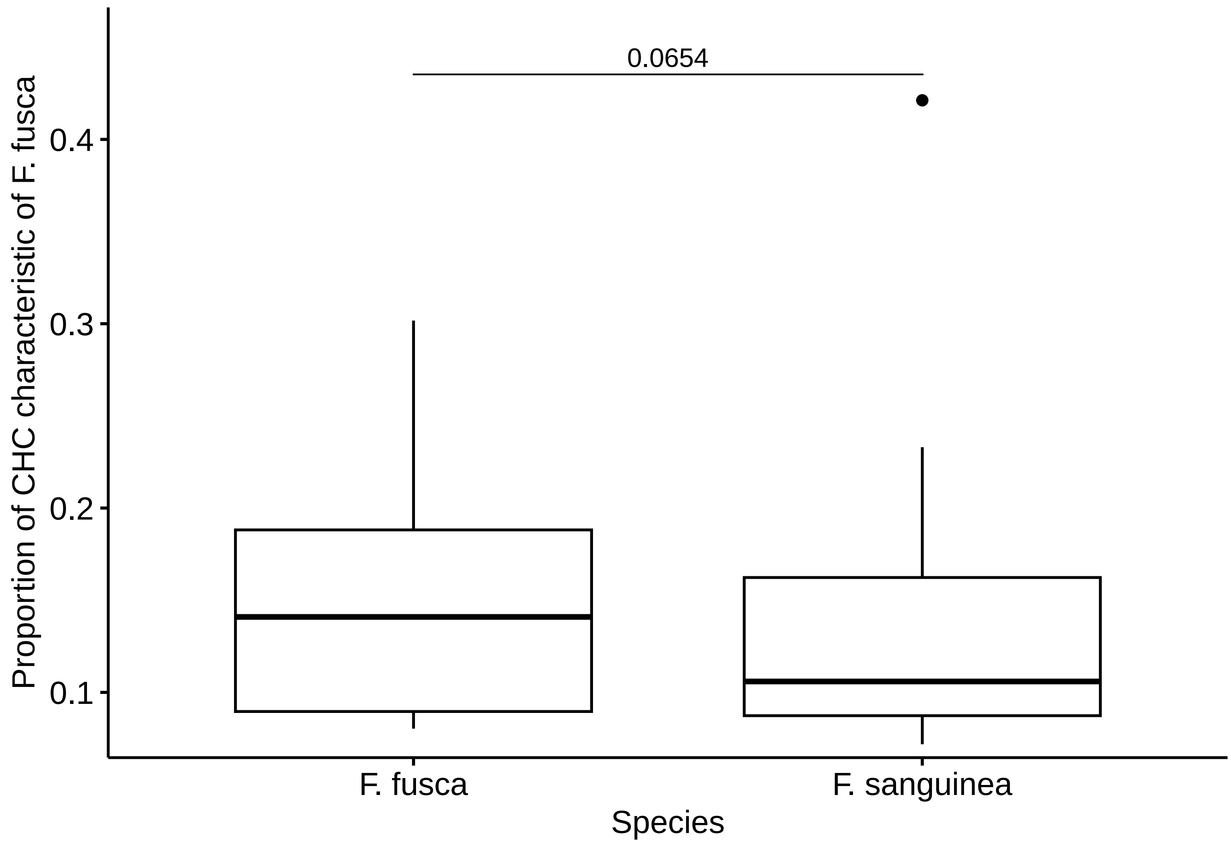
5.7 Difference in the proportion of CHC characteristic of *F. fusca* between mature *F. sanguinea* and *F. fusca*



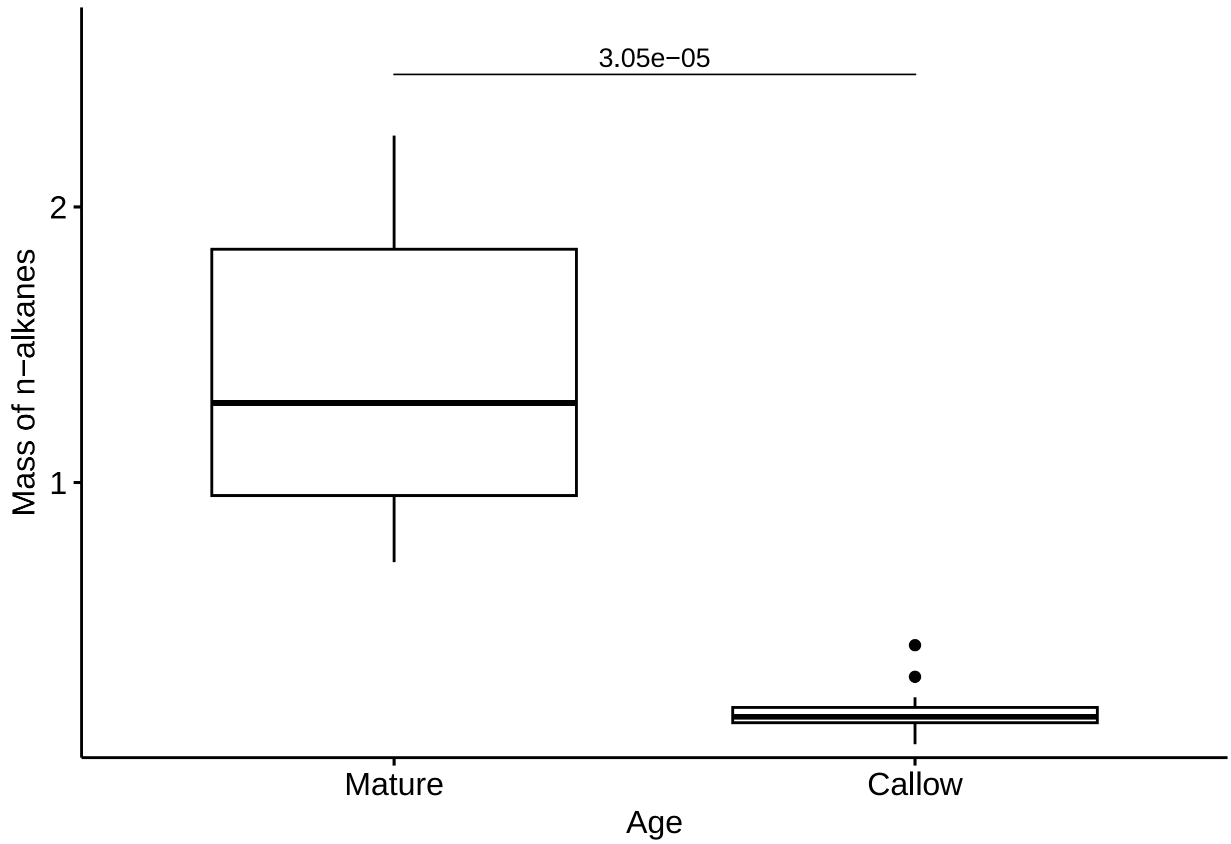
5.8 Difference in the proportion of CHC characteristic of *F. fusca* between mature and callow *F. sanguinea*



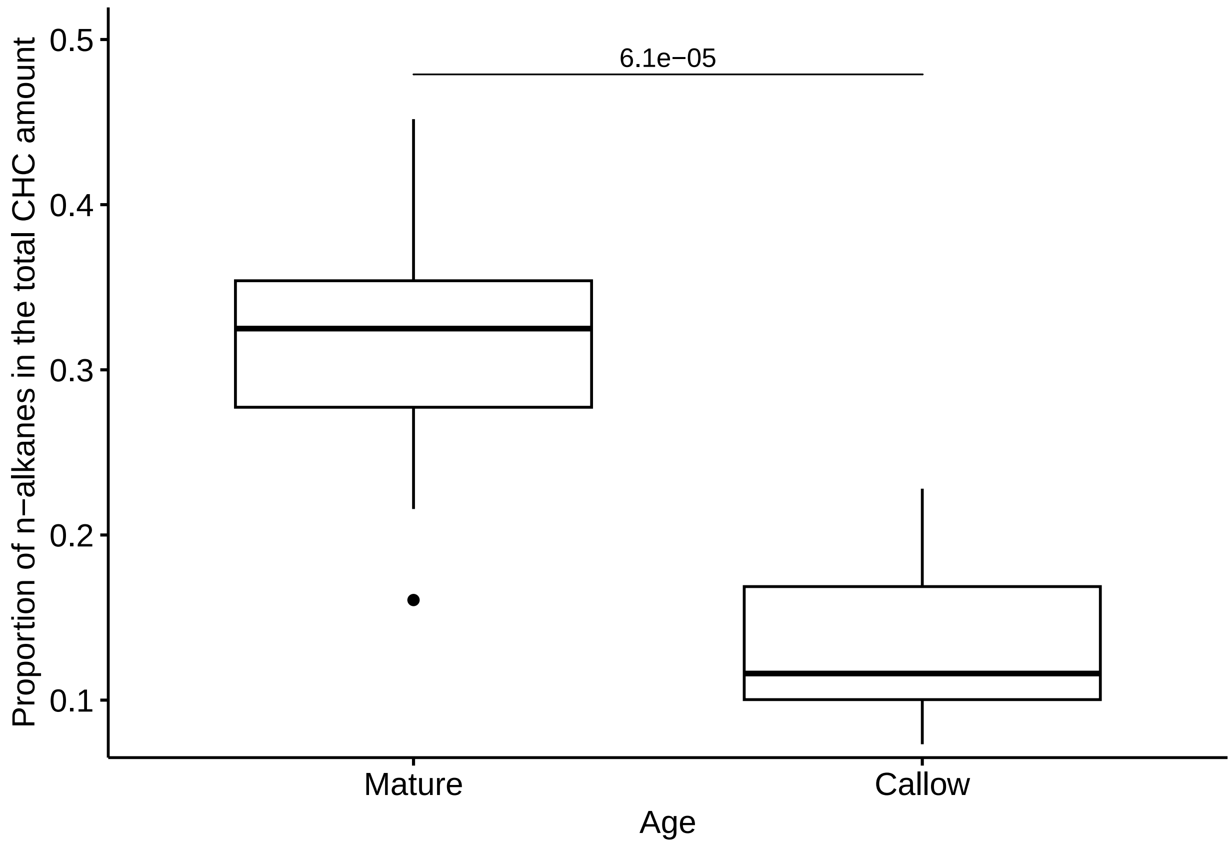
5.9 Difference in the proportion of CHC characteristic of *F. fusca* between callow *F. sanguinea* and *F. fusca*



5.10 Difference in the mass of *n*-alkanes between callow and mature *F. sanguinea*



5.11 Difference in the proprtion of *n*-alkanes between callow and mature *F. sanguinea*



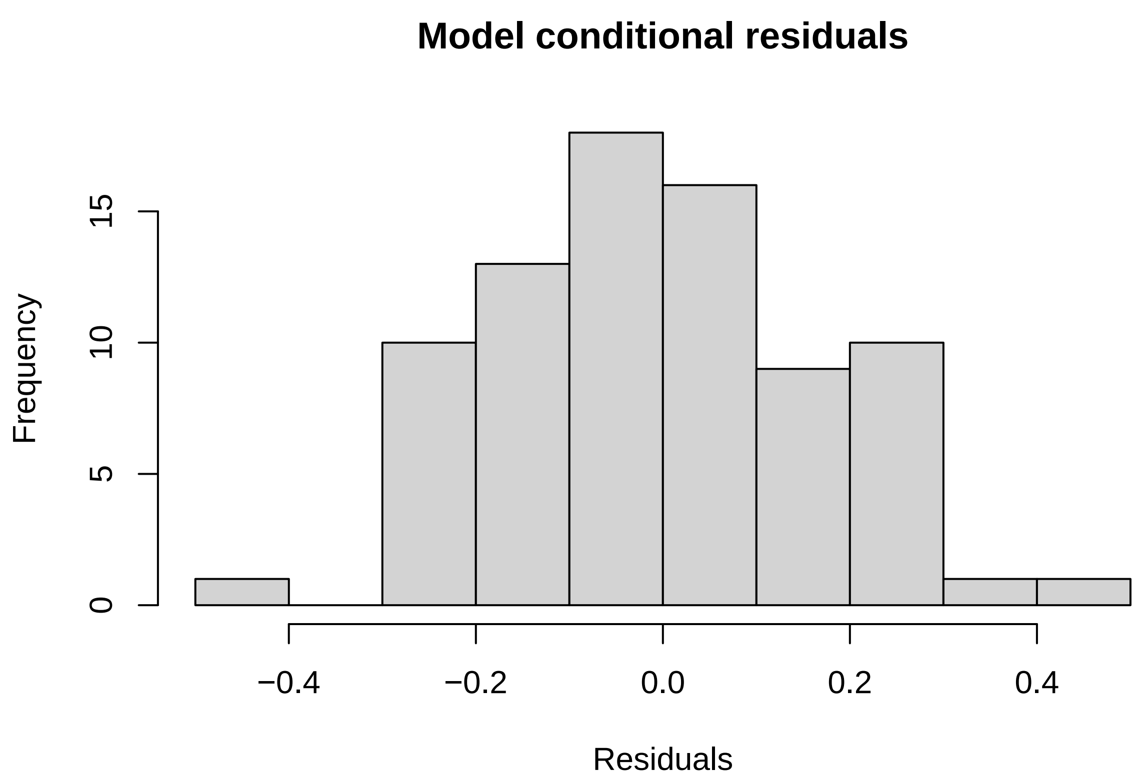
6 Change in the CHC profile of separated callow *F. sanguinea* ants

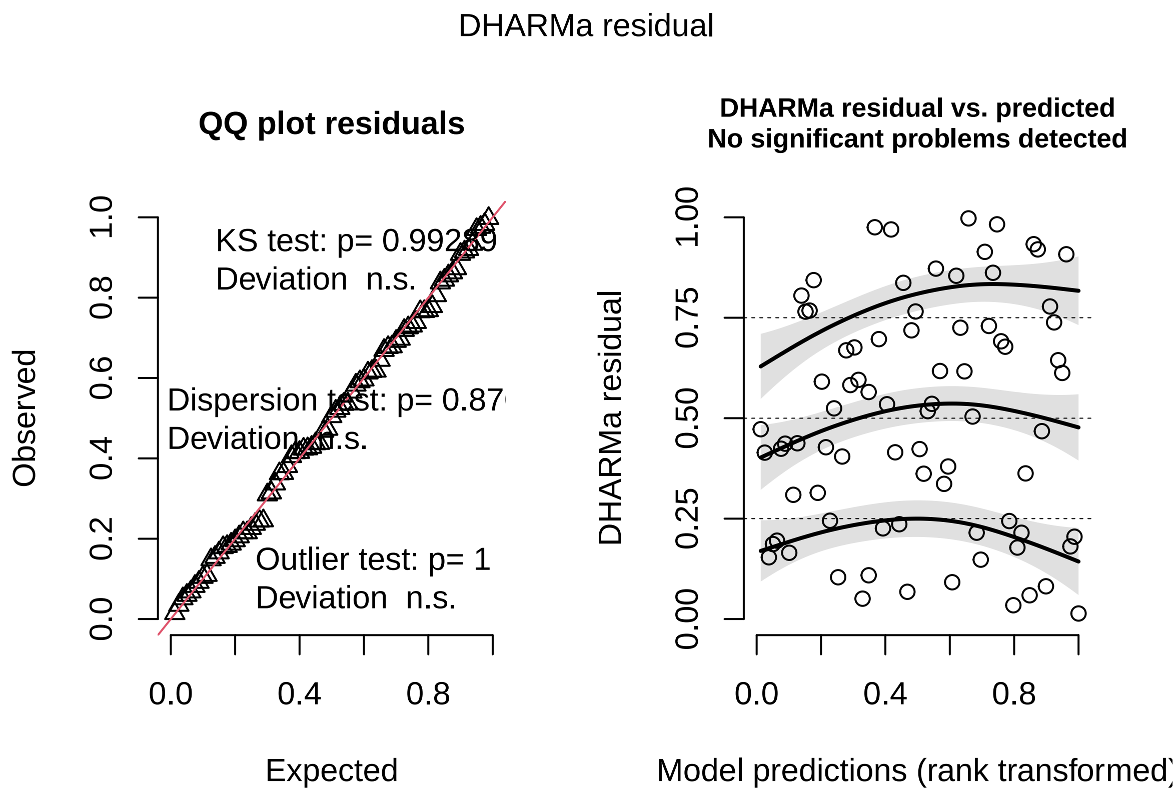
6.1 Change in the total CHC amount over time

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(corrected_mass)) ~ mean_delta + (1 | colony)
##    Data: separation_data[-74, ]
##
## REML criterion at convergence: -12.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.54596 -0.65494 -0.04106  0.63627  2.54062
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.01293  0.1137
##  Residual             0.03516  0.1875
## Number of obs: 79, groups:  colony, 11
```

```
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  1.180332   0.047605 16.450772  24.794 1.83e-14 ***
## mean_delta   0.002458   0.001669 68.538937   1.473    0.145
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##            (Intr)
## mean_delta -0.518
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99204, p-value = 0.9107
```



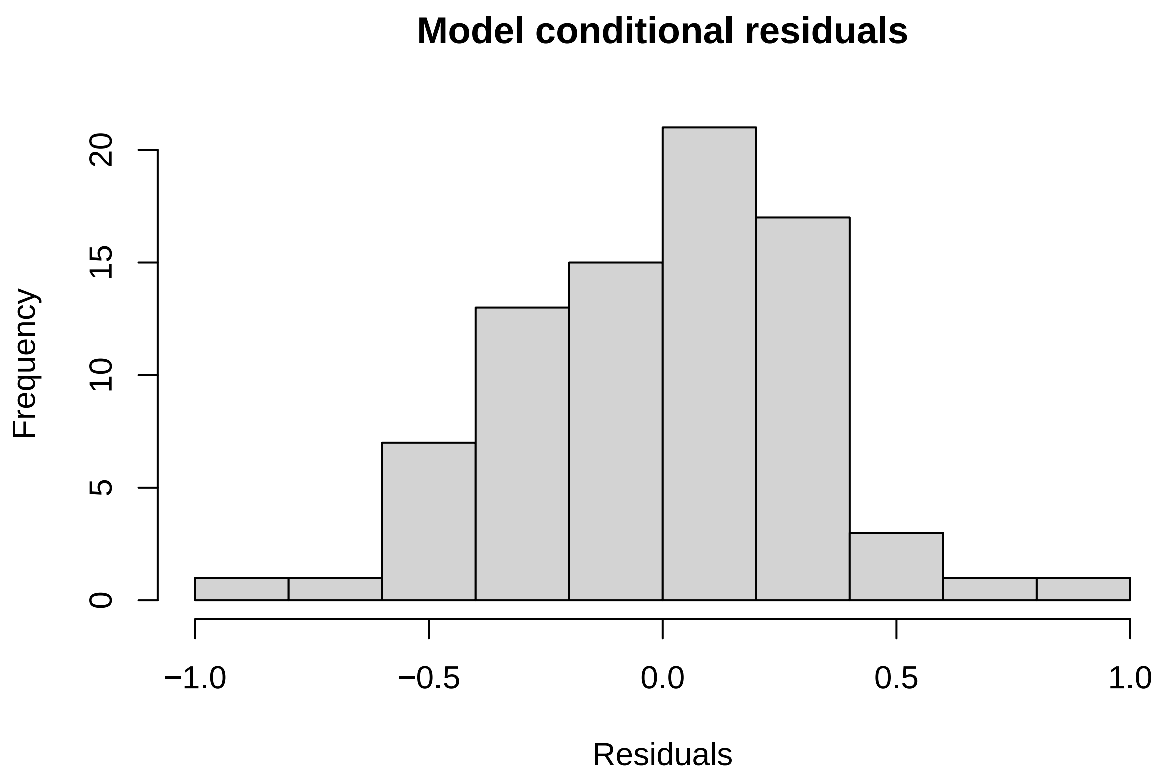


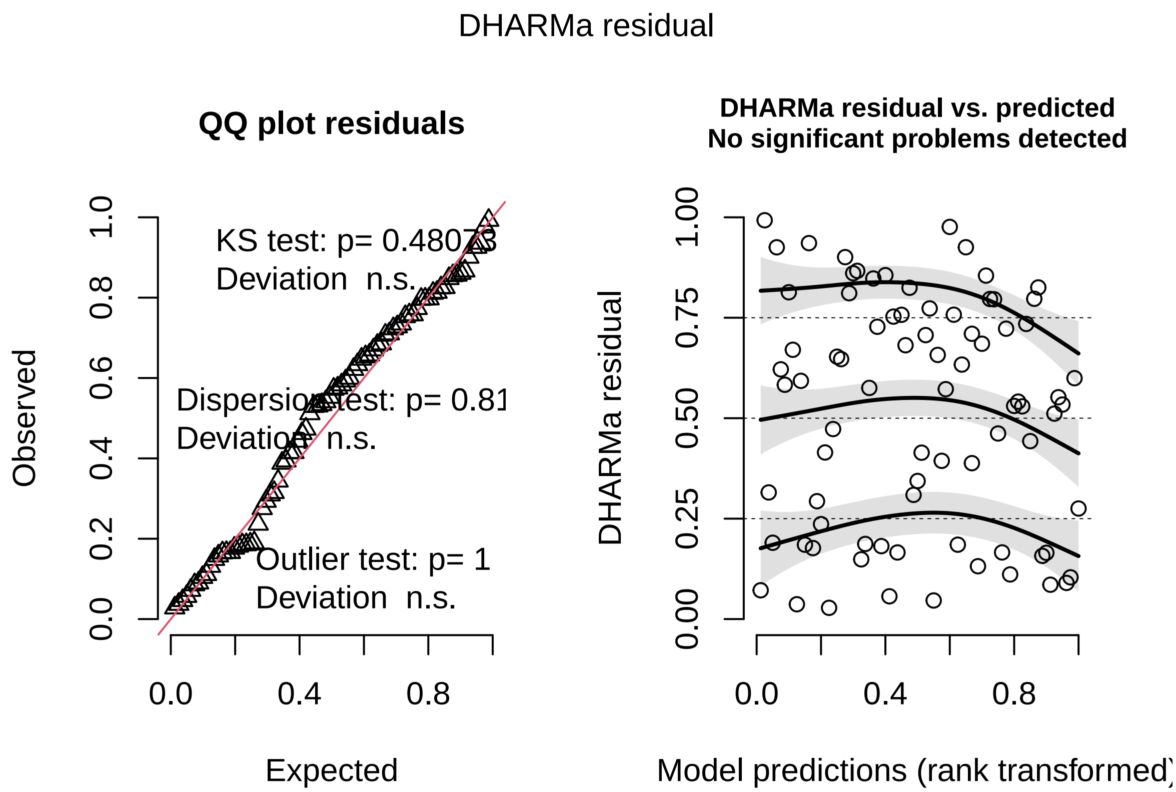
6.2 Change in the amount of compounds characteristic of callow *F. sanguinea*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: log(corrected_mass_part) ~ mean_delta + (1 | colony)
##    Data: separation_data
##
## REML criterion at convergence: 91.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.3229 -0.7788  0.1032  0.6331  2.5936
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.1122   0.3349
##  Residual             0.1227   0.3503
## Number of obs: 80, groups:  colony, 11
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept) -0.296830   0.118281 13.022925  -2.510   0.0261 *
## mean_delta  -0.004727   0.003041 68.343277  -1.554   0.1247
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##            (Intr)
## mean_delta -0.389

##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98968, p-value = 0.7747
```



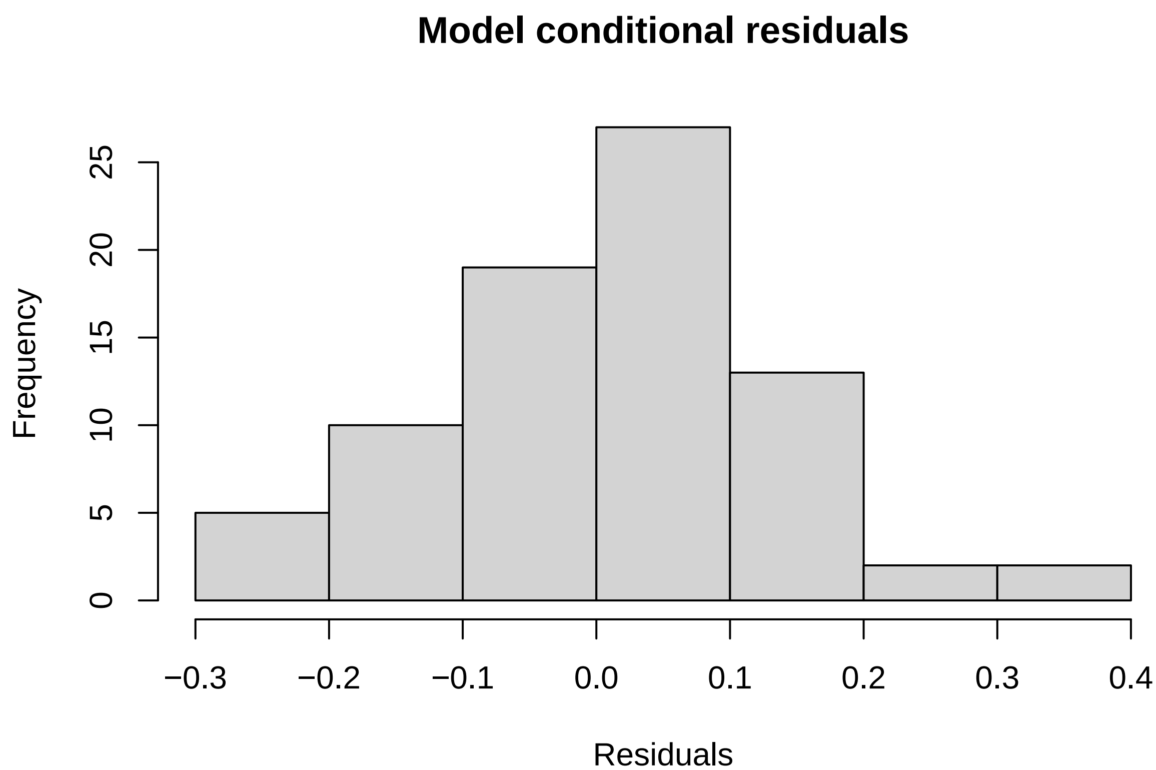


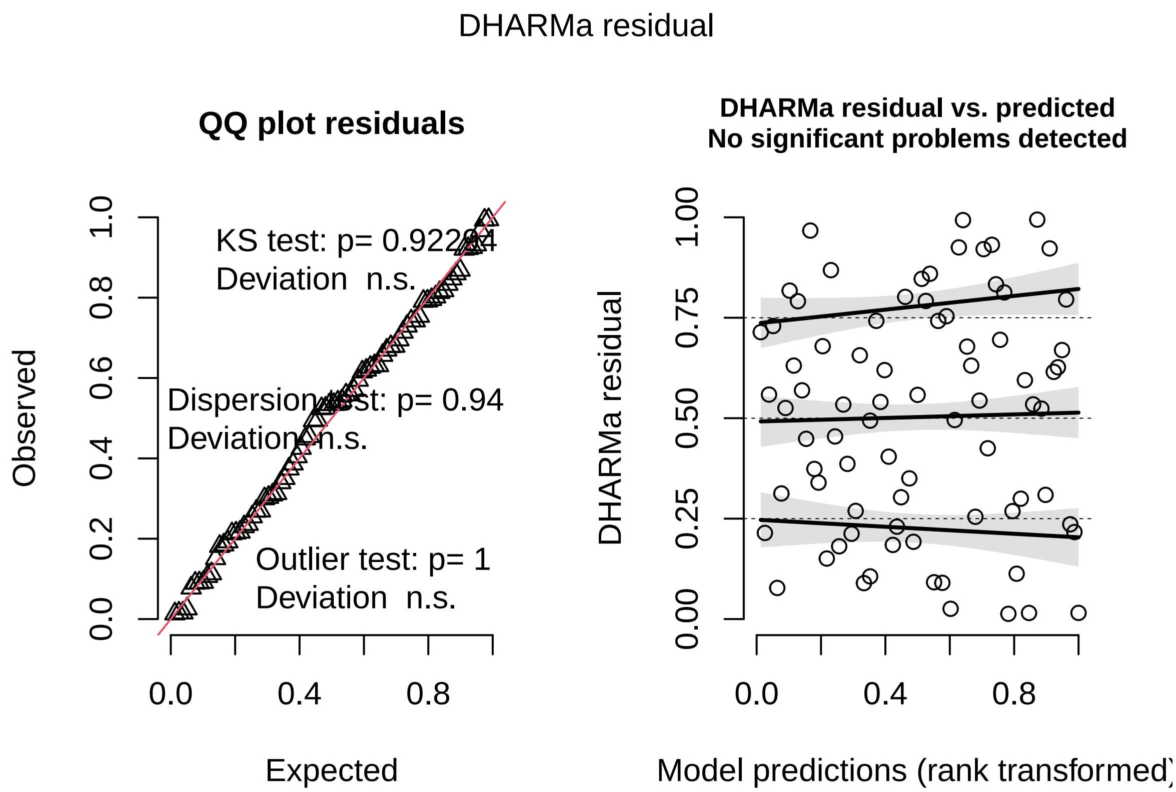
6.3 Change in the amount of compounds characteristic of *F. sanguinea*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(corrected_mass_part)) ~ poly(mean_delta, 1) + (1 | colony)
##    Data: separation_data[-c(69, 74), ]
##
## REML criterion at convergence: -89.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.24385 -0.62144  0.05051  0.71049  2.42988
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.001898 0.04356
##  Residual             0.015829 0.12581
## Number of obs: 78, groups:  colony, 11
##
## Fixed effects:
##                     Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)          0.52332    0.01968  7.96133  26.591 4.61e-09 ***
## poly(mean_delta, 1)  0.93404    0.12750 69.21774   7.326 3.35e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##             (Intr)
## ply(mn_d,1) 0.020

##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98807, p-value = 0.6851
```



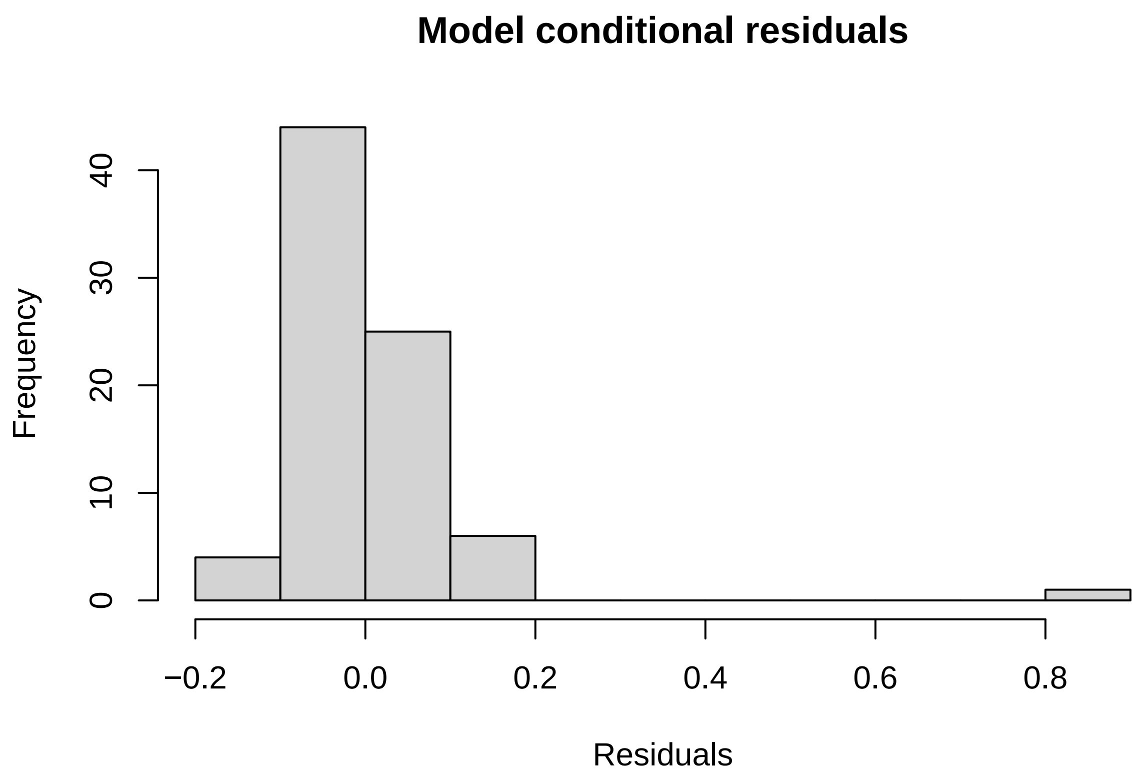


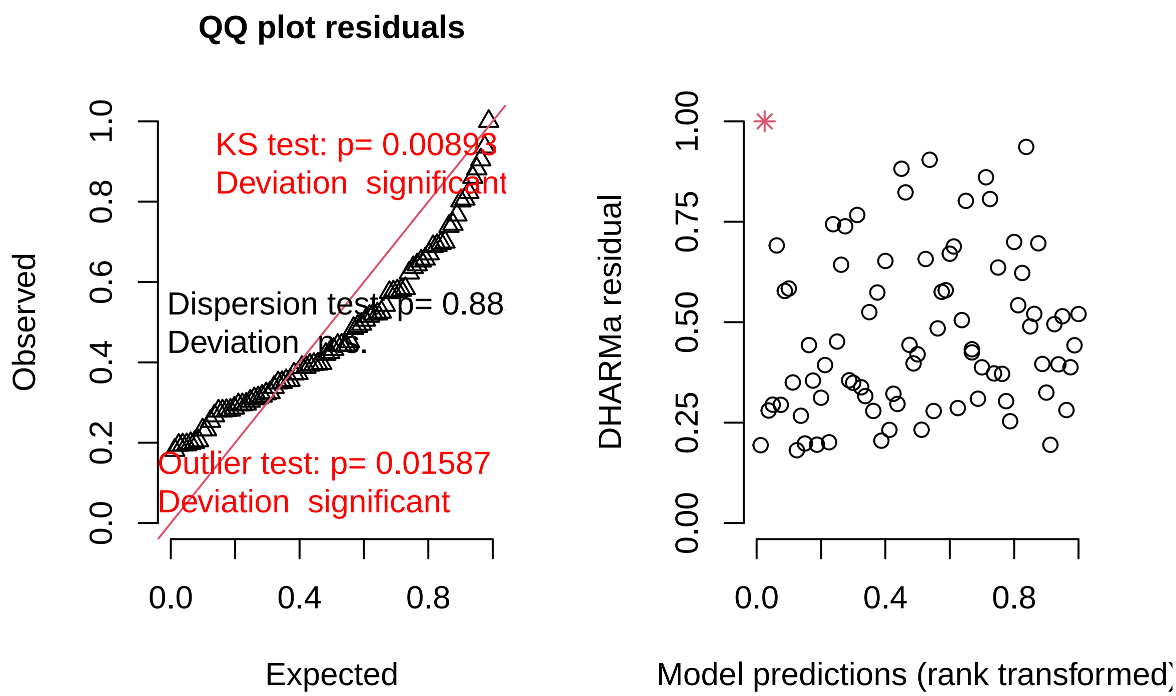
6.4 Change in the amount of compounds characteristic of *F. fusca*

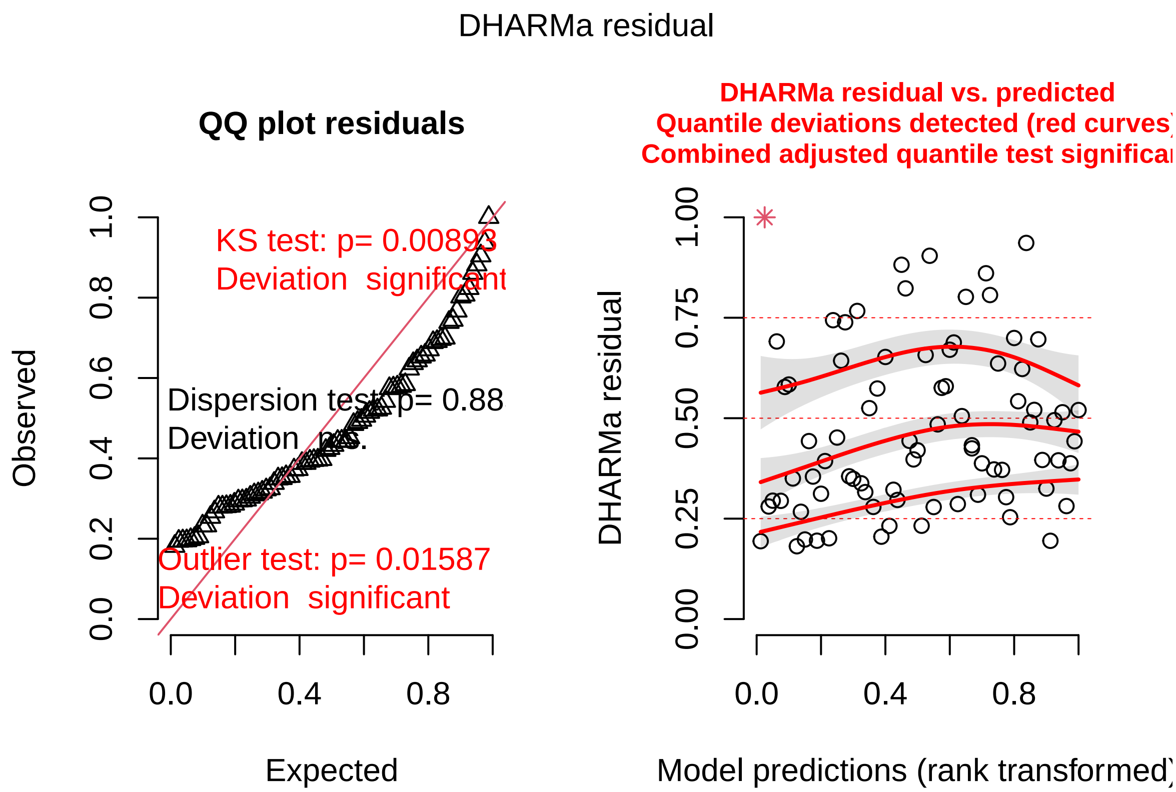
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: corrected_mass_part ~ mean_delta + (1 | colony)
##    Data: separation_data
##
## REML criterion at convergence: -72.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.5111 -0.4732 -0.1017  0.2072  6.8841
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.01203  0.1097
##  Residual             0.01524  0.1234
## Number of obs: 80, groups:  colony, 11
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)  1.669e-01  3.955e-02  1.386e+01   4.220 0.000875 ***
## mean_delta  -6.356e-05  1.071e-03  6.870e+01  -0.059 0.952851
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##            (Intr)
## mean_delta -0.410

##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.59749, p-value = 1.772e-13
```




QQ plot residuals
KS test: p= 0.00893
Deviation significant
Dispersion test: p= 0.88
Deviation n.s.
Outlier test: p= 0.01587
Deviation significant
Observed
Expected
DHARMa residual
Model predictions (rank transformed)



7 Impact of slaves on the development of the *F. sanguinea* CHC profile

We analyzed the impact of *F. fusca* slaves on the CHC profile of callow *F. sanguinea* ants. In this experiment, *F. sanguinea* ants were isolated in pairs before eclosion, and their CHC profiles were analyzed at one of four time points marking their adult age. We then tested whether callow *F. sanguinea* ants were chemically more similar to their slave relatives from free-living colonies. Unrelated *F. fusca* ants served as a background group."

7.1 Analysis based on all samples

```
## Empirical p-value for the chemical distance difference for the ants aged 1-3 days:0.00000

## Number of colonies used in the treatment of 1-3 days of separation: 9

## Empirical p-value for the chemical distance difference for the ants aged 8-10 days:0.00000

## Number of colonies used in the treatment of 8-10 days of separation: 9

## Empirical p-value for the chemical distance difference for the ants aged 17-20 days:0.00079

## Number of colonies used in the treatment of 17-20 days of separation: 9
```

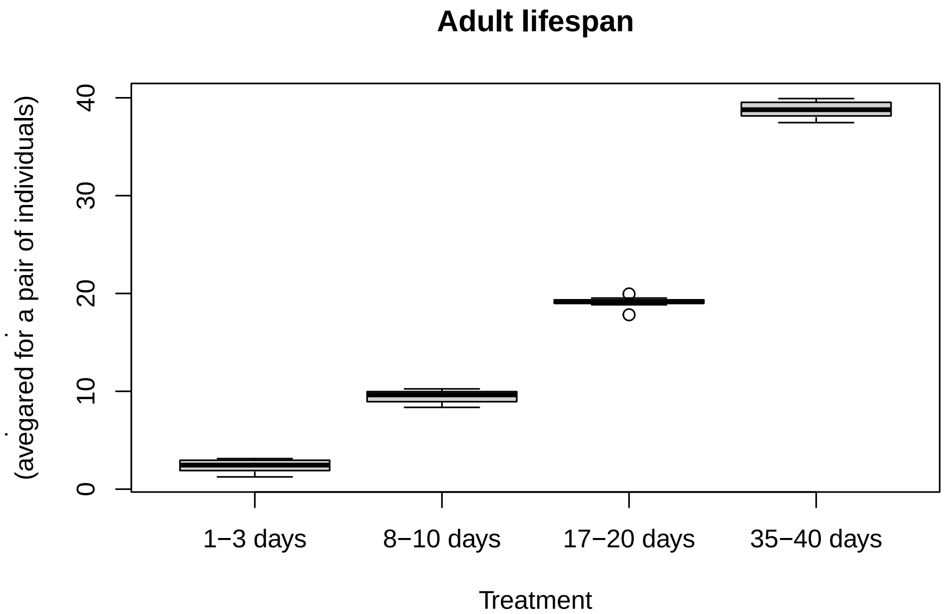


Figure S8 : Distribution of lifespans of ants used in the experiment. The precise age of individuals could not be determined because ants in pairs were not marked. Therefore, all possible combinations were considered to illustrate the potential range.

Table S3 : Chemical distance of the isolated *F. sanguinea* ant to *F. fusca* ants realted and unrelated to slaves.

Adult age	Chemical distance to slaves' relatives	Chemical distance to random colonies
1-3 days	0.3332	0.4718
8-10 days	0.3518	0.4802
17-20 days	0.4521	0.5291
35-40 days	0.5729	0.6100

```
## Empirical p-value for the chemical distance difference for the ants aged 35-40 days:0.00703
```

```
## Number of colonies used in the treatment of 35-40 days of separation: 8
```

7.2 Analysis restricted to the individuals hatched from cocoons

Since naked pupae might have acquired CHC through physical contact with adult ants after pupation, we subset our data to include only those samples which derived from the ants hatched from cocoons. In this case silky envelop during pupa stage ruled out possibility of the CHC transfer from the environment.

```
## Empirical p-value for the chemical distance difference for the ants aged 1-3 days:0.00000
```

```
## Number of colonies used in the treatment of 1-3 days of separation: 8
```

```
## Empirical p-value for the chemical distance difference for the ants aged 8-10 days:0.00001
```

Table S4 : Chemical distance of the isolated *F. sanguinea* ant to *F. fusca* ants realted and unrelated to slaves. Isolated ant spun a silky envelope before pupal stage, which prevented CHC transfer from the enviroment.

Adult age	Chemical distance to slaves' relatives	Chemical distance to random colonies
1-3 days	0.3473	0.4869
8-10 days	0.3740	0.4932
17-20 days	0.4643	0.5407
35-40 days	0.6175	0.6458

```
## Number of colonies used in the treatment of 8-10 days of separation: 8

## Empirical p-value for the chemical distance difference for the ants aged 17-20 days:0.00438

## Number of colonies used in the treatment of 17-20 days of separation: 8

## Empirical p-value for the chemical distance difference for the ants aged 35-40 days:0.03298

## Number of colonies used in the treatment of 35-40 days of separation: 7
```

8 Effect of dummy ants

In these experiments, *F. sanguinea* ants released from their cocoon envelopes were maintained in Petri dishes along with glass beads coated with CHC extracted from either *F. fusca* or *F. sanguinea* ants. We examined whether the contact with dummy nestmates influenced the development of CHC profile of young *F. sanguinea* workers. In particular, we aimed to determine there there is evidence for active chemical mimicry in *F. sanguinea*.

8.1 Distance to the CHC profile of the dummy ants treated by *F. fusca* CHC

We computed the chemical distance to the CHC profile of ants whose CHC were used to cover glass beads serving as dummy ants. In the control variant, we used the ants subjected to clean glass beads.

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  to_test$mean_dist.x and to_test$mean_dist.y
## V = 18.5, p-value = 0.6781
## alternative hypothesis: true location shift is not equal to 0
```

8.2 Distance to the CHC profile of the dummy ants treated by *F. sanguinea* CHC

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  to_test$mean_dist.x and to_test$mean_dist.y
## V = 12, p-value = 0.2361
## alternative hypothesis: true location shift is not equal to 0
```

Table S5 : Chemical distance of separated *F. sanguinea* ants to the CHC profile of *F. fusca* ants from colonies that served as a source of CHC to caot the glass beads. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged distance	Contrast	Averaged distance
SD18-3	12-15 days + F. fusca hydrocarbons	0.396	12-15 days (control)	0.509
SD18-5	12-15 days + F. fusca hydrocarbons	0.616	12-15 days (control)	0.513
SD18-5	12-15 days + F. fusca hydrocarbons	0.842	12-15 days (control)	0.832
SD19-11	12-15 days + F. fusca hydrocarbons	0.467	12-15 days (control)	0.469
SD19-3	12-15 days + F. fusca hydrocarbons	0.500	12-15 days (control)	0.436
SD19-4	12-15 days + F. fusca hydrocarbons	0.305	12-15 days (control)	0.424
SD19-6	12-15 days + F. fusca hydrocarbons	0.674	12-15 days (control)	0.699
SD19-8	12-15 days + F. fusca hydrocarbons	0.595	12-15 days (control)	0.587
SD20-2	12-15 days + F. fusca hydrocarbons	0.699	12-15 days (control)	0.709

Table S6 : Chemical distance of separated *F. sanguinea* ants to the CHC profile of *F. sanguinea* ants from colonies that served as a source of CHC to caot the glass beads. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged distance	Contrast	Averaged distance
SD18-2	12-15 days + F. sanguinea hydrocarbons	0.571	12-15 days (control)	0.373
SD18-3	12-15 days + F. sanguinea hydrocarbons	0.457	12-15 days (control)	0.599
SD18-5	12-15 days + F. sanguinea hydrocarbons	0.525	12-15 days (control)	0.580
SD18-5	12-15 days + F. sanguinea hydrocarbons	0.250	12-15 days (control)	0.232
SD19-11	12-15 days + F. sanguinea hydrocarbons	0.595	12-15 days (control)	0.607
SD19-3	12-15 days + F. sanguinea hydrocarbons	0.419	12-15 days (control)	0.511
SD19-4	12-15 days + F. sanguinea hydrocarbons	0.648	12-15 days (control)	0.668
SD19-6	12-15 days + F. sanguinea hydrocarbons	0.553	12-15 days (control)	0.591
SD19-8	12-15 days + F. sanguinea hydrocarbons	0.850	12-15 days (control)	0.850
SD20-2	12-15 days + F. sanguinea hydrocarbons	0.646	12-15 days (control)	0.654

Table S7 : Proportion of *n*-docosane in CHC extracted from *F. sanguinea* ants maintained with the glass beads coated with the CHC of *F. fusca* ants and contaminated with *n*-docosane. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged C22 fraction	Contrast	Averaged C22 fraction
SD18-3	12-15 days + F. fusca hydrocarbons	0.018	12-15 days (control)	0.000
SD18-5	12-15 days + F. fusca hydrocarbons	0.001	12-15 days (control)	0.000
SD18-5	12-15 days + F. fusca hydrocarbons	0.002	12-15 days (control)	0.000
SD19-11	12-15 days + F. fusca hydrocarbons	0.011	12-15 days (control)	0.002
SD19-3	12-15 days + F. fusca hydrocarbons	0.009	12-15 days (control)	0.001
SD19-4	12-15 days + F. fusca hydrocarbons	0.017	12-15 days (control)	0.000
SD19-6	12-15 days + F. fusca hydrocarbons	0.009	12-15 days (control)	0.001
SD19-8	12-15 days + F. fusca hydrocarbons	0.003	12-15 days (control)	0.002
SD20-2	12-15 days + F. fusca hydrocarbons	0.009	12-15 days (control)	0.005
W17-1	12-15 days + F. fusca hydrocarbons	0.021	12-15 days (control)	0.001

8.3 Fraction of *n*-docosane in the CHC profile

For the control treatment, glass beads were left uncoated with CHC. In the experimental treatments, the CHC used to coat the glass beads were supplemented with *n*-docosane, which served as an internal standard. We examined differences in the proportion of *n*-docosane in the CHC profiles of the tested ants, as these could indicate the acquisition of chemicals from the dummy ants.

8.3.1 *F. fusca*

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  to_test$C22_prop.x and to_test$C22_prop.y
## V = 55, p-value = 0.005857
## alternative hypothesis: true location shift is not equal to 0
```

8.3.2 *F. sanguinea*

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  to_test$C22_prop.x and to_test$C22_prop.y
## V = 54, p-value = 0.007093
## alternative hypothesis: true location shift is not equal to 0
```

Table S8 : Proportion of *n*-docosane in CHC extracted from *F. sanguinea* ants maintained with the glass beads coated with the CHC of *F. sanguinea* ants and contaminated with *n*-docosane. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged C22 fraction	Contrast	Averaged C22 fraction
SD18-2	12-15 days + F. sanguinea hydrocarbons	0.005	12-15 days (control)	0.002
SD18-3	12-15 days + F. sanguinea hydrocarbons	0.014	12-15 days (control)	0.000

SD18-5	12-15 days + F. sanguinea hydrocarbons	0.003	12-15 days (control)	0.000
SD18-5	12-15 days + F. sanguinea hydrocarbons	0.002	12-15 days (control)	0.000
SD19-11	12-15 days + F. sanguinea hydrocarbons	0.004	12-15 days (control)	0.002
SD19-3	12-15 days + F. sanguinea hydrocarbons	0.003	12-15 days (control)	0.001
SD19-4	12-15 days + F. sanguinea hydrocarbons	0.002	12-15 days (control)	0.000
SD19-6	12-15 days + F. sanguinea hydrocarbons	0.004	12-15 days (control)	0.001
SD19-8	12-15 days + F. sanguinea hydrocarbons	0.002	12-15 days (control)	0.002
SD20-2	12-15 days + F. sanguinea hydrocarbons	0.004	12-15 days (control)	0.005
W17-1	12-15 days + F. sanguinea hydrocarbons	0.003	12-15 days (control)	0.001

9 Body surface area of *F. sanguinea* ants and their slaves

In the publication, CHC amounts are reported relative to the square of head width. To assess whether this measure reliably reflects body size, we measured the planar projection areas of different body parts of *F. sanguinea* and *F. fusca*. The plots below suggest that head width slightly underestimates body surface area in *F. fusca* compared to *F. sanguinea*. As a result, the relative concentration of CHC in *F. fusca* may be overestimated. However, this bias acts in the opposite direction of our findings—*F. sanguinea* exhibits greater CHC concentration on its cuticle. Therefore, the shape-related bias between species is conservative, as it reduces the likelihood of rejecting the null hypothesis

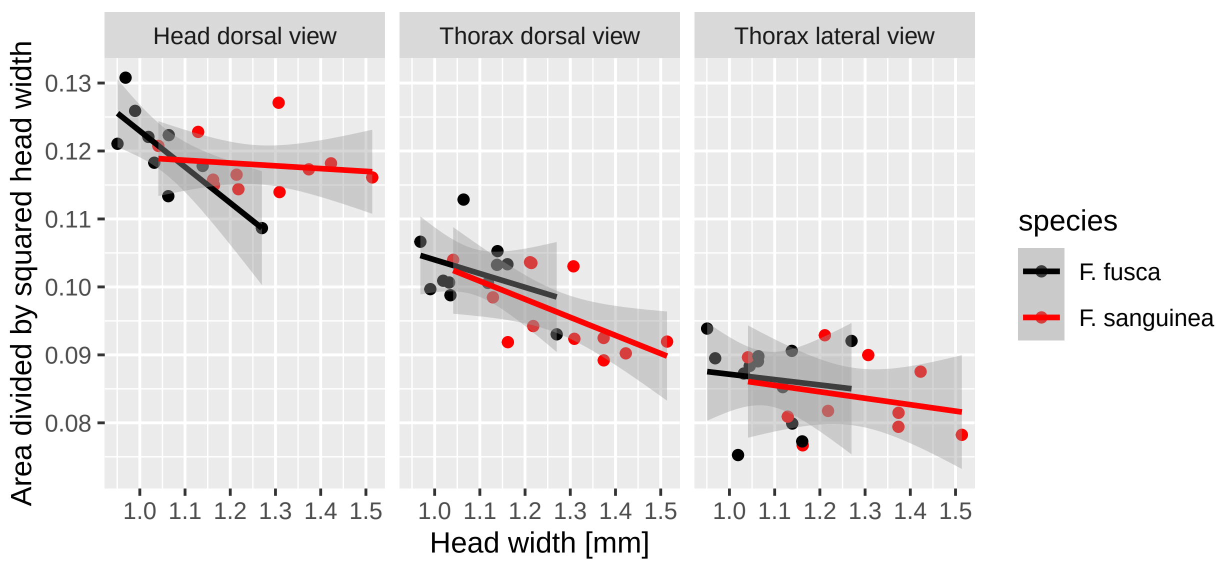


Figure S9 : Ratios of the planar projection areas of three body parts (head, thorax dorsal view, thorax lateral view) to the square of head width.

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